

Endogenous Firm Scope via Organizational Complexity*

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Abstract

This paper develops a theory of endogenous firm scope in which the binding constraint on expansion is organizational complexity—the cumulative coordination burden generated by heterogeneity across activities—rather than the number of markets served. Firms operate a non-rival organizational blueprint that can be deployed across markets, but effective deployment requires scarce headquarters attention for market-specific adaptation. As the firm’s footprint grows more heterogeneous, organizational congestion raises the shadow cost of attention throughout the organization, making expansion decisions inherently non-separable. The theory delivers sharp predictions: optimal scope follows a cutoff rule in organizational distance from the firm’s core, generating related diversification and a core–periphery internal structure. When organizational technology is sufficiently strong, a phase transition occurs—constraints cease to bind, scope becomes unbounded, and platform-type firms emerge as a distinct organizational regime. The same mechanism that bounds scope also compresses the right tail of the firm size distribution: larger firms face more heterogeneous marginal markets, so differences in capability translate into progressively smaller differences in scope. Organizational complexity thus provides a unified explanation for why most firms remain bounded, why platforms emerge discontinuously, and why very large firms are scarce.

Keywords: Firm scope; organizational complexity; intangible capital; heterogeneous markets; non-rivalry; platform firms

JEL Codes: D23; E22; L22; L23; O32; O33

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Large modern firms operate across an extraordinarily wide range of markets, products, and activities. Digital platforms such as Amazon, Google, or Uber simultaneously serve millions of distinct customer–product–location combinations, apparently at negligible marginal organizational cost. Yet many firms of comparable productivity and resources remain narrowly focused, and even the largest platforms exhibit sharp boundaries: Amazon dominates e-commerce logistics but has not replicated this success in healthcare; Google spans digital advertising and cloud computing but struggles in hardware. A central challenge for the theory of the firm is therefore not merely to explain why some firms become large, but to explain how firms select which activities to span when potential markets differ in their organizational requirements—and why scope appears cheaply expandable in some directions but prohibitively costly in others.

Existing theories of firm size and scope typically impose limits through fixed managerial spans (Lucas, 1978), discrete hierarchies of knowledge and control (Garicano, 2000; Garicano and Rossi-Hansberg, 2006), or per-market entry costs (Melitz, 2003). A common feature of these frameworks is that expansion decisions are effectively separable across markets or tasks: serving one activity does not alter the organizational cost of serving another, except through aggregate resource constraints. As a result, scope is mechanically bounded by technological or informational limits that scale with the number of activities undertaken, rather than by endogenous interactions across heterogeneous activities. This paper proposes a different view. We argue that the binding constraint on firm scope is not the count of markets served, but the cumulative organizational complexity generated by heterogeneity across activities. Firms can scale across vast numbers of highly similar markets at low marginal cost, while expanding into even a small number of sufficiently distinct environments sharply strains internal coordination. Organizational complexity—formalized as the integral of market heterogeneity over the firm’s footprint—becomes the scarce resource that disciplines scope, induces non-separability across expansion decisions, and endogenously determines which markets the firm serves.

To formalize these ideas, we develop a model in which firms operate a centralized, non-rival blueprint that can be deployed across a continuum of heterogeneous markets. The blueprint—representing codified organizational knowledge, operational procedures, or proprietary technology—is created at headquarters and can be used simultaneously in any market the firm chooses to serve. However, effective deployment of the blueprint in a given market requires market-specific adaptation using scarce headquarters attention. Markets differ in their organizational distance from the firm’s core: some are nearly identical to the environment in which the blueprint was developed, while others require substantial customization. We summarize this vari-

ation through a heterogeneity index for each market and define organizational complexity as the integral of heterogeneity over the firm's footprint. Complexity enters the firm's problem through an organizational congestion function that raises the shadow price of headquarters attention as the firm's footprint becomes more heterogeneous. The firm therefore faces a fundamental trade-off: the blueprint is non-rival and can be deployed at zero marginal cost in terms of replication, but each market served adds to organizational complexity and tightens the constraint on scarce headquarters resources. Scope is thus neither free nor mechanically costly—it depends on which markets the firm selects. This framework delivers four sets of results.

The first set of results establishes that optimal firm scope is bounded in organizational complexity rather than market count. Even though firms may operate across an arbitrarily large collection of activities, the total heterogeneity of those activities must remain finite at the optimum—otherwise, organizational congestion would drive the shadow cost of headquarters attention to infinity. This result reconciles two observations that appear contradictory under count-based theories: firms can scale across vast numbers of highly similar markets (as platforms do), yet the mass of materially distinct environments any firm serves remains sharply limited. Endogenous limits to scope emerge without imposing fixed per-market costs or discrete managerial constraints.

The second set of results characterizes a scope–technology tradeoff that governs expansion decisions. Holding blueprint quality fixed, adding markets raises organizational complexity and tightens internal congestion, reducing the value of incumbent activities. At the same time, because the blueprint is non-rival, serving more markets increases the marginal return to improving blueprint quality. These two forces operate in opposite directions: scope expansion simultaneously taxes existing operations and strengthens incentives to invest in organizational technology. Firm scope is therefore determined by the balance of these forces rather than by separable market-level profitability, and the profitability of any given market depends on the composition of the firm's entire footprint.

The third set of results concerns market selection and internal organization. When markets can be ordered by heterogeneity, optimal scope exhibits a cutoff structure: firms serve all markets sufficiently close to their organizational core and exclude those beyond a critical distance. This selection rule provides a theoretical foundation for the empirical finding that scope expansion occurs predominantly along related dimensions. Within the chosen footprint, scarce headquarters attention is allocated unevenly—concentrated in low-heterogeneity core activities where adaptation is most productive, while peripheral activities are operated through more standardized processes. A core–periphery structure thus emerges endogenously from organizational scarcity, even in the

absence of technological asymmetries or fixed costs.

The fourth set of results characterizes platform emergence as a phase transition in organizational regime. Allowing firms to invest in standardization technology does not, by itself, eliminate organizational limits: when blueprint quality is low, the firm remains in a bounded-scope regime where complexity constraints bind. However, when blueprint quality exceeds a critical threshold and near-core market opportunities are sufficiently abundant, firms can replicate a common organizational recipe across arbitrarily many markets while keeping total complexity finite. Beyond this threshold, platform-scale operation becomes not only feasible but optimal, and internal organizational constraints cease to bind at the margin. Platform firms therefore emerge as a distinct organizational form—not merely large firms, but firms that have crossed a technological boundary at which the economics of scope fundamentally change.

Finally, we trace out implications of the organizational complexity mechanism for the cross-sectional distribution of firm scope. Because the marginal market for a larger firm is more heterogeneous than the marginal market for a smaller firm, the mapping from underlying productivity to scope is concave: differences in capability translate into progressively smaller differences in scope as firms grow. When firm productivity follows a Pareto distribution—as standard models assume and evidence supports—the distribution of firm scope exhibits a thinner right tail than the productivity distribution that generates it. Organizational complexity thus provides a structural explanation for the relative scarcity of very large firms: the “missing giants” in the right tail of the firm size distribution reflect not the absence of highly capable firms, but the organizational constraints that prevent capability from translating fully into scope.

Related literature This paper contributes to three distinct literatures on firm organization, scope, and scale. First, we build on the span-of-control and organizational economics tradition initiated by [Lucas \(1978\)](#) and developed by [Garicano \(2000\)](#) and [Garicano and Rossi-Hansberg \(2006\)](#). In these models, firm scope is mechanically bounded by discrete managerial or knowledge constraints: each additional activity requires a fixed or irreducible quantum of supervisory attention, so the number of tasks a firm can undertake is finite. Related work on coordination and information-processing costs ([Radner, 1993](#); [Zandt, 1999](#)) emphasizes congestion in decentralized decision-making. Our approach departs from this tradition by modeling organizational complexity as a continuous object that aggregates heterogeneity across activities rather than counting them. Scope is not limited by a fixed managerial span but by an endogenous organizational constraint that firms can relax through investment in blueprint quality and adaptation technology. This formulation allows firms to operate across an unbounded measure of sufficiently similar markets while

serving only a finite mass of materially distinct ones—a distinction that count-based models cannot capture.

Second, we contribute to the literature on intangible capital and firm scope. This literature builds on the insight that organizational knowledge is non-rival within the firm. [McGrattan and Prescott \(2010\)](#) formalize this property in the context of multinational production, showing that intangible capital can be deployed across locations without being depleted. Subsequent work extends this logic: [Argente, Moreira, Oberfeld, and Venkateswaran \(2025\)](#) show that standardization lowers marginal costs and facilitates smooth expansion, while [Crouzet, Eberly, Eisfeldt, and Papanikolaou \(2024\)](#) emphasize replicability and external contracting frictions. In the trade literature, [Bernard, Redding, and Schott \(2010, 2011\)](#) develop models of multi-product firms with a core competence, where productivity declines with distance from that core. A common feature of these frameworks is that scope decisions remain separable: each market or product is evaluated independently, and serving one activity does not alter the organizational cost of serving another. Our paper shares the non-rivalry property of intangible capital but introduces a countervailing force: deploying the blueprint across heterogeneous markets requires adaptation using scarce headquarters attention, and the organizational burden aggregates across the firm’s footprint. This non-separability—absent from the existing literature—is what generates the selection problem, the core–periphery structure, the phase transition to platform-type organization, and the compression of the firm size distribution.

Third, our framework complements the empirical literature on management practices and firm performance ([Bloom and Van Reenen, 2007](#); [Bloom, Sadun, and Van Reenen, 2019](#)) by providing a structural link between management technology and firm scope. In our model, improved management practices operate through more effective blueprint creation and adaptation efficiency, allowing firms to compress heterogeneity across markets and expand scope discontinuously. This connects to the platform literature ([Bresnahan and Greenstein, 1999](#); [Rochet and Tirole, 2003](#); [Gawer and Cusumano, 2014](#)), but through a distinct channel: platform emergence in our framework arises from the endogenous collapse of internal organizational constraints rather than from external network effects. The core–periphery structure that emerges within the firm resonates with evidence that large organizations concentrate managerial oversight in core activities while operating peripheral lines through standardized processes ([Antràs and Rossi-Hansberg, 2009](#)).

1 Environment

We model a single firm that operates across a set of heterogeneous markets. The firm is organized around a centralized, non-rival blueprint created at headquarters and deployed across markets. Effective deployment of the blueprint in a given market requires market-specific adaptation. Serving markets generates an organizational burden at headquarters, summarized by a scalar state variable called organizational complexity, which depends on the composition of the firm’s footprint rather than on the number of markets served. The remainder of this section formalizes these elements.

1.1 Markets and Heterogeneity

We formalize markets and heterogeneity through four distinct objects, each building on the previous:

1. A market space (S, Σ, μ) providing the measurable structure over which firms optimize.
2. Market characteristics $\xi(s) \in \mathbb{R}^k$ capturing the primitive, multidimensional sources of heterogeneity at each market $s \in S$.
3. A heterogeneity index $d(s) \in \mathbb{R}_+$ aggregating characteristics into a scalar measure of organizational difficulty.
4. Organizational complexity $C(M) \in \mathbb{R}_+$ aggregating heterogeneity across the firm’s footprint into a firm-level state variable.

This layered structure separates the abstract space of markets from the economic content of heterogeneity, and separates market-level heterogeneity from its firm-level consequences.

Layer 1: Market space. Let (S, Σ, μ) be a measure space of markets, indexed by $s \in S$. This space is the primitive object carrying all measurable structure; the σ -algebra Σ and measure μ determine which collections of markets are measurable and how they are weighted. A distinguished origination location H (“headquarters”) lies outside S ; no goods production takes place at H , and all production occurs in markets $s \in S$.

We use the term **market** broadly. An index s may represent:

- a geographic market,
- a product line,
- a business unit,
- a customer segment,
- or any operational activity requiring blueprint deployment.

Assumption 1.1 (Non-atomic market space). *The measure space (S, Σ, μ) is non-atomic: for every $E \in \Sigma$ with $\mu(E) > 0$ and every $\varepsilon \in (0, \mu(E))$, there exists $E' \subseteq E$ with $E' \in \Sigma$ and $\mu(E') = \varepsilon$.*

Assumption 1.1 rules out indivisible markets of positive measure and ensures that any collection of markets can be subdivided arbitrarily finely. This is natural in environments where markets differ continuously along multiple characteristics and is standard in models with continua of activities.

Layer 2: Market characteristics. Each market $s \in S$ is characterized by a vector of heterogeneity-relevant attributes

$$\xi(s) \in \mathbb{R}^k,$$

where $\xi : S \rightarrow \mathbb{R}^k$ is measurable with respect to Σ and the Borel σ -algebra on $\mathbb{R}^k$. The components of $\xi(s)$ capture distinct primitive sources of market heterogeneity—geography, technology, regulation, customer composition, operational requirements, and so forth. These characteristics are exogenous to the firm.

In addition, each market s has local rival factor capacity $\bar{X}(s) > 0$, also exogenous.

Layer 3: Heterogeneity index. We summarize the multidimensional characteristics $\xi(s)$ through a scalar heterogeneity index

$$d(s) = \delta(\xi(s)),$$

where $\delta : \mathbb{R}^k \rightarrow \mathbb{R}_+$ is Borel measurable. The index $d(s)$ measures organizational difficulty: markets with higher $d(s)$ require weakly more adaptation effort to achieve a given level of effective blueprint deployment.

Assumption 1.2 (Nonnegativity). *The heterogeneity index satisfies $d(s) \geq 0$ for μ -a.e. $s \in S$.*

Example 1.3 (Organizational distance). *A useful interpretation of $d(s)$ is as organizational distance from the firm's core. For instance, $d(s)$ may take the form of a weighted norm,*

$$d(s) = \left(\sum_{j=1}^k \omega_j |\xi_j(s)|^p \right)^{1/p},$$

or an additive index $d(s) = \sum_{j=1}^k d_j(\xi_j(s))$. Different specifications correspond to different organizational technologies and substitution patterns across dimensions of heterogeneity. None of the results require $d(\cdot)$ to satisfy the axioms of a norm.

Layer 4: Organizational complexity. Serving markets imposes an organizational burden at headquarters. We summarize this burden by a scalar state variable called organizational complexity, which aggregates market-level heterogeneity across the firm’s entire footprint.

Definition 1 (Organizational complexity). *For any measurable set of served markets $M \subseteq S$, organizational complexity is*

$$C(M) \equiv \int_M d(s) \mu(ds). \quad (1)$$

Organizational complexity $C(M)$ is well defined in $[0, \infty]$ for any measurable $M \subseteq S$. By construction, $C(\cdot)$ satisfies the following properties for any measurable sets $M_1, M_2 \subseteq S$:

- *Nonnegativity:* $C(M) \geq 0$.
- *Monotonicity:* if $M_1 \subseteq M_2$, then $C(M_1) \leq C(M_2)$.
- *Additivity:* if $M_1 \cap M_2 = \emptyset$, then $C(M_1 \cup M_2) = C(M_1) + C(M_2)$.

Remark 1.4 (Role of the layered structure). *The four-layer structure separates the abstract mathematical framework from its empirical implementation. Layers 1 and 4 define the theoretical objects—the market space (S, Σ, μ) and organizational complexity $C(M)$ —while Layers 2 and 3 provide the bridge to data. In applications, the econometrician observes market characteristics $\xi(s)$ and must specify or estimate the aggregator $\delta(\cdot)$ that maps these characteristics into organizational heterogeneity $d(s) = \delta(\xi(s))$. The theoretical results hold for any measurable $d : S \rightarrow \mathbb{R}_+$, providing flexibility in how this mapping is constructed. This separation ensures that the theoretical propositions remain valid across a wide range of empirical settings, while the layered structure provides clear guidance on where modeling choices and data requirements enter.*

1.2 Blueprint Technology

Blueprint. The firm operates a centralized blueprint, denoted by $z \geq 0$. The blueprint is a non-rival organizational asset created at headquarters with the following properties:

- *Non-rivalry:* a given blueprint z can be deployed simultaneously across all served markets.
- *Centralization:* the blueprint is created and maintained at headquarters.
- *Genericity:* effective deployment of the blueprint in market s requires market-specific adaptation.

The blueprint does not directly produce output. Instead, it enters market-level payoffs through its effective deployment, which depends on local conditions and organizational inputs defined below.

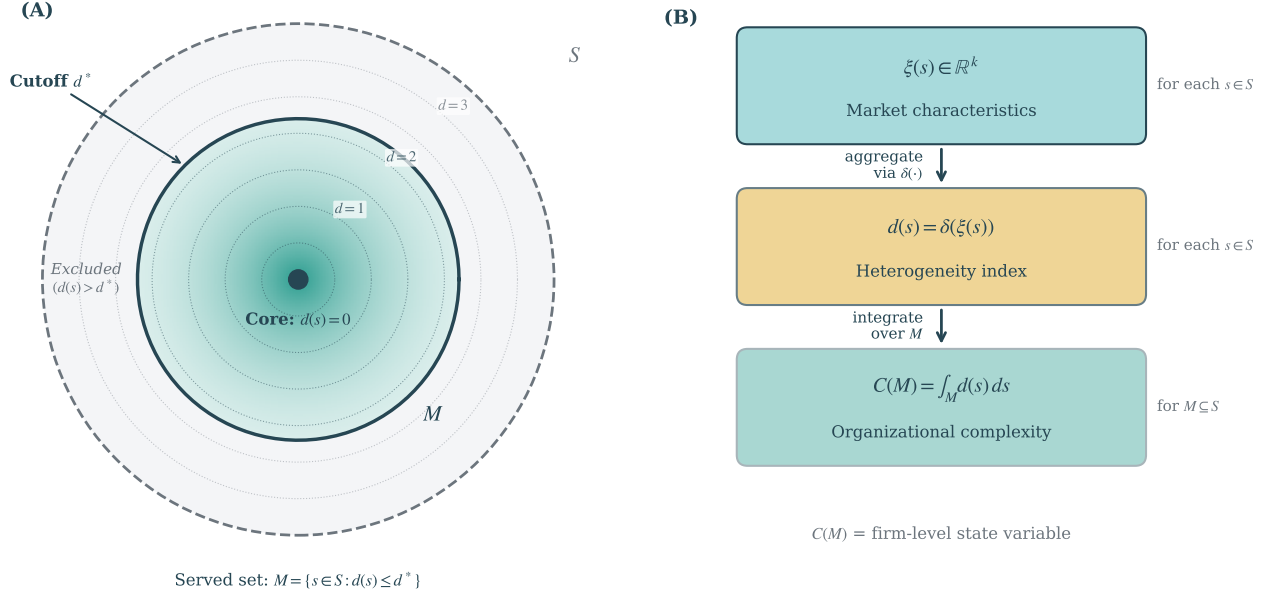


Figure 1: Market heterogeneity and endogenous firm scope.

Notes. Panel (a) depicts the market space S ordered by heterogeneity and the firm's endogenous footprint $M \subset S$. Under optimal organization, the firm serves all markets with heterogeneity below a cutoff d^* , so that $M = \{s \in S : d(s) \leq d^*\}$, while more heterogeneous markets are excluded. Panel (b) illustrates the layered structure of heterogeneity. Each market $s \in S$ is characterized by multidimensional attributes $\xi(s) \in \mathbb{R}^k$, which are aggregated into a scalar heterogeneity index $d(s) = \delta(\xi(s))$. Organizational complexity is defined as the integral of heterogeneity over the served set, $C(M) = \int_M d(s) ds$.

Blueprint creation. The firm produces a blueprint at headquarters using headquarters attention. Let $h_z \geq 0$ denote the amount of headquarters attention allocated to blueprint creation. Blueprint quality is given by

$$z = \Gamma(h_z), \quad (2)$$

where $\Gamma : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is strictly increasing and continuous with $\Gamma(0) = 0$. Since Γ is strictly increasing, it admits an inverse $\chi \equiv \Gamma^{-1}$, so that producing a blueprint of quality z requires

$$h_z = \chi(z). \quad (3)$$

Blueprint deployment and adaptation. To deploy the blueprint in market s , the firm adapts it to local conditions using both headquarters and local organizational inputs. Let $h(s) \geq 0$ denote headquarters attention allocated to market s , and let $\ell(s) \geq 0$ denote local adaptation effort in market s . We restrict attention to measurable allocations $h(\cdot)$ and $\ell(\cdot)$.

Effective adaptation in market s is given by

$$a(s) = A(h(s), \ell(s)), \quad (4)$$

and effective blueprint deployment in market s is

$$z_s = \theta(a(s), d(s)) z, \quad (5)$$

where $\theta(\cdot, \cdot) \in (0, 1]$ is a portability function. The heterogeneity index $d(s)$ enters (10) as the sufficient statistic for how the primitive characteristics $\xi(s)$ affect blueprint portability.

Assumption 1.5 (Adaptation and portability). *The function $A : \mathbb{R}_+^2 \rightarrow \mathbb{R}_+$ is continuous and weakly increasing in each argument. The portability function $\theta : \mathbb{R}_+ \times \mathbb{R}_+ \rightarrow [0, 1]$ is continuous, weakly increasing in a , weakly decreasing in d , and satisfies*

$$\lim_{a \rightarrow \infty} \theta(a, d) = 1 \quad \text{for all } d \geq 0.$$

Assumption 1.5 captures that blueprint deployment requires market-specific adaptation using both headquarters and local inputs, and that adaptation improves the effectiveness with which the blueprint can be applied. Greater heterogeneity reduces portability, but sufficiently intensive adaptation can fully overcome these frictions.

1.3 Organizational Constraints and Payoffs

Local rival capacity. Each market s has limited local organizational resources $\bar{X}(s) > 0$. These resources must be allocated between production and local adaptation. Let $x(s) \geq 0$ denote production input in market s . Local feasibility requires

$$x(s) + \ell(s) \leq \bar{X}(s). \quad (6)$$

Scarcity of headquarters attention. Headquarters attention is rival within the firm and must be allocated across competing uses: blueprint creation and market-specific adaptation. We model organizational scarcity through a complexity-dependent coordination wedge: as the firm serves a more heterogeneous set of markets, each unit of headquarters attention becomes more costly to

deploy. Deploying headquarters attention then incurs the total cost

$$q(C(M)) \left(\chi(z) + \int_M h(s) \mu(ds) \right). \quad (7)$$

Assumption 1.6 (Organizational congestion). *The organizational congestion function $q : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ satisfies:*

1. $q(0) = 1$;
2. q is continuously differentiable and strictly increasing, with $q'(C) > 0$ for all $C \geq 0$;
3. $\lim_{C \rightarrow \infty} q(C) = \infty$.

We refer to $q(C)$ as organizational congestion. It captures the idea that broader and more heterogeneous scope raises coordination burdens, thereby increasing the marginal cost of headquarters attention anywhere in the firm. We normalize the base price of headquarters attention to 1 (in units of the numeraire), so $q(C)$ represents the complexity-adjusted shadow price of attention.

Market-level payoff. Operating in market s generates payoff

$$R(z_s, x(s)), \quad (8)$$

where $R(\cdot, \cdot)$ is weakly increasing in both arguments. The function R summarizes demand, pricing, and production conditions in market s . We assume $s \mapsto R(z_s, x(s))$ is measurable for all feasible allocations and impose Assumption 1.7 to ensure that the firm's objective is well defined.

Assumption 1.7 (Conditional integrability). *For any fixed blueprint quality $z \geq 0$, there exists an integrable function $\bar{R}_z \in L^1(S, \mu)$ such that for all measurable $M \subseteq S$ and all feasible allocations $\{x(s), \ell(s), h(s)\}_{s \in M}$ consistent with (10)–(11),*

$$R(z_s, x(s)) \leq \bar{R}_z(s) \quad \text{for } \mu\text{-a.e. } s \in S.$$

Assumption 1.7 requires that, for any fixed blueprint quality z , the firm's total potential gross payoff from serving markets is finite. The assumption does not impose a uniform bound on returns across different blueprint qualities and therefore does not restrict the scale or profitability of technological improvements. Instead, it rules out pathological cases in which, holding the firm's technology fixed, payoffs could be made arbitrarily large by serving an unbounded set of markets or by choosing extreme local allocations.

1.4 Firm Problem

The firm chooses a blueprint quality $z \geq 0$, a measurable set of markets $M \subseteq S$, and measurable allocations $\{x(s), \ell(s), h(s)\}_{s \in M}$ to maximize net payoff:

$$\max_{z, M, \{x, \ell, h\}} \left\{ \int_M R(z_s, x(s)) \mu(ds) - q(C(M)) \left(\chi(z) + \int_M h(s) \mu(ds) \right) \right\}, \quad (9)$$

subject to, for all $s \in M$,

$$z_s = \theta(A(h(s), \ell(s)), d(s)) z, \quad (10)$$

$$x(s) + \ell(s) \leq \bar{X}(s). \quad (11)$$

Served markets. We interpret the firm's footprint M as the support of market-level activity. Formally, a market is served if the firm allocates positive resources to it:

$$s \in M \iff x(s) + \ell(s) + h(s) > 0.$$

Accordingly, whenever we refer to a feasible plan $(M, z, \{x, \ell, h\})$, we take $M = \{s \in S : x(s) + \ell(s) + h(s) > 0\}$ up to μ -null sets.

2 Baseline Results

The results characterize how firms organize their scope and internal operations in the presence of organizational complexity. We distinguish between two margins. The extensive margin determines which markets are served and governs the firm's overall footprint. The intensive margin determines how organizational resources are allocated across served markets. Both margins are disciplined by the same underlying mechanism—organizational complexity and the scarcity of headquarters attention—but they operate in distinct ways. The results below first establish limits and ordering on the extensive margin, and then characterize how firms optimally allocate headquarters effort across markets on the intensive margin.

2.1 Organizational Complexity and Endogenous Limits to Firm Scope

We need several assumptions first.

Assumption 2.1 (Positive heterogeneity). $d(s) > 0$ for μ -a.e. $s \in S$.

Assumption 2.1 rules out a positive-measure set of perfectly frictionless markets with zero heterogeneity. It ensures that any non-negligible expansion of scope entails some organizational burden, while still allowing the firm to scale across many nearly homogeneous markets with arbitrarily small $d(s)$.

Assumption 2.2 (No free deployment). *For any market s with $d(s) > 0$, blueprint portability satisfies*

$$\theta(A(0, \ell), d(s)) = 0 \quad \text{for all } \ell \geq 0.$$

Assumption 2.2 rules out costless deployment of the blueprint across heterogeneous markets, even when the firm invests arbitrarily in blueprint scale. As a result, serving any market with positive heterogeneity requires strictly positive organizational resources, preventing free expansion along the extensive margin. Local expertise cannot perfectly substitute for headquarters attention; some HQ involvement is always required to deploy the blueprint.

Assumption 2.3 (No positive payoff without effective deployment). *For any market s and any feasible local condition $x \in \mathcal{X}_s$ implied by feasibility constraint (11), the payoff satisfies*

$$R(0, x) \leq 0.$$

Assumption 2.3 rules out free-standing market rents: favorable local market conditions alone cannot generate positive payoffs for the firm in the absence of effective blueprint deployment.

We are in a position to state the first formal result of the paper.

Proposition 2.1 (Bounded organizational complexity). *Suppose that organizational congestion is unbounded (Assumption 1.6), that gross payoffs are conditionally integrable (Assumption 1.7), and that positive payoffs require effective blueprint deployment (Assumptions 2.2 and 2.3). Then any optimal firm choice $(M^*, z^*, \{x^*, \ell^*, h^*\})$ satisfies*

$$C(M^*) = \int_{M^*} d(s) \mu(ds) < \infty.$$

Proof. See Appendix A.1. □

Proposition 2.1 formalizes a bounded organizational capacity view of scope: even though the firm may operate across a very large collection of markets, the total organizational burden generated by heterogeneity, $C(M) = \int_M d(s) \mu(ds)$, must remain finite at the optimum because escalating complexity raises the marginal cost of headquarters resources through $q(C)$.

Corollary 2.1 (Finite mass of materially distinct markets). *Fix any heterogeneity threshold $\tau > 0$ and define the set of materially distinct markets*

$$M_\tau^* \equiv \{s \in M^* : d(s) \geq \tau\}.$$

Then

$$\mu(M_\tau^*) \leq \frac{1}{\tau} \int_{M^*} d(s) \mu(ds) = \frac{C(M^*)}{\tau} < \infty,$$

so the optimal organization serves only a finite (in measure) set of markets whose heterogeneity exceeds τ .

Proof. See Appendix A.1. □

Corollary 2.1 translates Proposition 2.1 into an economically meaningful statement about the mass of materially distinct markets. For any resolution level $\tau > 0$, the firm can serve only a finite (in measure) set of markets with heterogeneity at least τ . At the same time, the model allows the firm to cover a vast—and potentially unbounded—measure of highly similar markets with $d(s) < \tau$, and in particular with arbitrarily small $d(s)$.

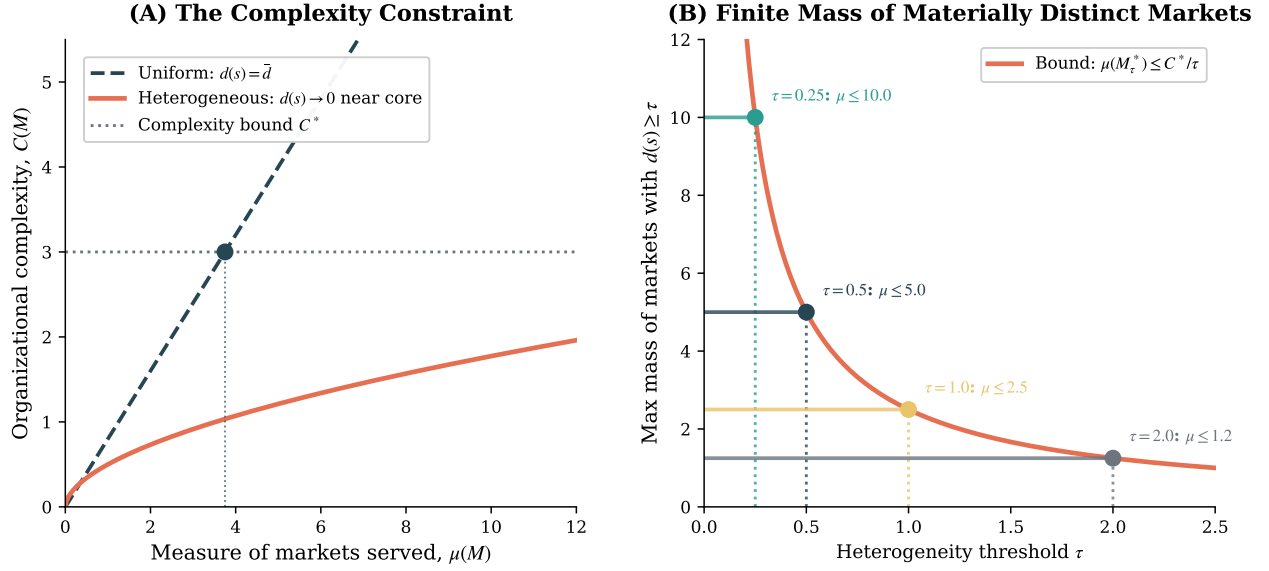


Figure 2: Organizational complexity and bounded scope

This distinction captures the scalability of platform-like firms. For example, Amazon can serve millions of product–location niches because most of them are handled through standardized logistics, modular IT systems, and templated compliance processes—in the model, these niches correspond to markets with very small heterogeneity $d(s)$. The binding constraint is not the raw count of niches, but the aggregate heterogeneity of the footprint: the mass of materially different environments (large- d markets) that require bespoke adaptation and managerial attention.

Relation to span-of-control theories. Classic theories of firm size and scope based on span of control, such as [Lucas \(1978\)](#) and [Garicano \(2000\)](#), impose an exogenous limit on expansion through a fixed or irreducible managerial burden associated with each additional activity or task. In these models, managerial attention is scarce and expansion requires discrete allocations of supervisory or problem-solving resources. As a result, firm scope is mechanically bounded: the number of markets or activities a firm can operate in is finite and determined by technological parameters governing managerial capacity.

The mechanism in our model is fundamentally different. We do not impose per-market fixed costs, discrete entry costs, or uniform lower bounds on organizational burden. Instead, the constraint on expansion operates through total organizational complexity,

$$C(M) = \int_M d(s) \mu(ds),$$

which aggregates heterogeneity across markets in a smooth and continuous manner. This formulation allows firms to serve a large—potentially unbounded—set of sufficiently similar markets, while sharply limiting the mass of materially distinct environments that require bespoke adaptation and coordination.

As a consequence, firm scope is genuinely endogenous. Firms expand not by exhausting a fixed span of managerial control, but by endogenously reshaping the heterogeneity associated with additional markets through standardization, modular organizational design, and scalable operational blueprints. In this sense, large platform-type firms—such as Amazon—operate across vast numbers of markets not because managerial constraints are absent, but because organizational heterogeneity has been compressed, allowing scope to scale without proportionate increases in coordination costs. This distinction separates our approach from traditional span-of-control models and highlights a novel channel through which organizational technology governs firm scope.

2.2 Organizational Complexity, Scale, and the Scope Trade-off

Definition 2 (Market-level value function). *Fix a market $s \in S$, a blueprint quality $z \geq 0$, and an organizational complexity level $C \geq 0$. The market-level value function $\Pi_s(z; C)$ is defined as the value of the inner optimization problem*

$$\begin{aligned} \Pi_s(z; C) &\equiv \sup_{\substack{x \geq 0, \ell \geq 0, h \geq 0 \\ x + \ell \leq \bar{X}(s)}} \left\{ R(z_s, x) - q(C) h \right\}, \\ \text{s.t.} \quad z_s &= \theta(A(h, \ell), d(s)) z. \end{aligned} \tag{12}$$

The blueprint creation cost $q(C)\chi(z)$ is common across markets and is handled at the outer problem.

The function $\Pi_s(z; C)$ represents the value of operating market s given the organizational prices induced by complexity C and a fixed blueprint quality z , taking the firm's overall scope and blueprint investment as given.

Assumption 2.4 (Blueprint effectiveness). *For each market s , the market-level payoff function $R(z_s, x)$ is weakly increasing in effective blueprint deployment z_s .*

Assumption 2.4 formalizes that greater effective deployment of the blueprint weakly improves performance in any given market. The assumption imposes no curvature or diminishing returns on market payoffs and serves only to ensure that blueprint investment is economically meaningful rather than a purely organizational artifact.

Proposition 2.2 (Organizational congestion). *Fix a market $s \in S$ and a blueprint level $z \geq 0$. The market-level value function $\Pi_s(z; C)$ is weakly decreasing in organizational complexity C . In particular, for any served set $M \subseteq S$ and any measurable $\Delta \subseteq S \setminus M$ with $\mu(\Delta) > 0$,*

$$\Pi_s(z; C(M \cup \Delta)) \leq \Pi_s(z; C(M)).$$

Proof. See Appendix A.2. □

Remark 2.5 (Strictness). *The inequality in Proposition 2.2 is strict for an incumbent market $s \in M$ whenever the optimal inner allocation at $C(M)$ uses strictly positive headquarters attention, $h^*(s; z, C(M)) > 0$, so that the shadow price $q(C)$ is locally relevant. Markets operated purely through the standardized blueprint ($h^*(s) = 0$) are locally insulated from changes in organizational complexity.*

Proposition 2.2 isolates a negative internal organizational externality that operates conditional on blueprint quality. Holding the blueprint z fixed, expanding the firm's footprint raises organizational complexity and thereby increases the shadow price of headquarters attention. Because headquarters attention is a shared and rival input used for market-level adaptation, this increase in organizational scarcity weakly lowers the value of all incumbent markets.

This mechanism implies that market selection is inherently non-separable. The profitability of operating in a given market depends not only on local conditions, but also on the composition of the firm's overall footprint through its effect on organizational complexity. Importantly, this does not imply that expansion is unprofitable in aggregate: new markets may generate sufficient value to offset the organizational tax imposed on incumbent activities. Rather, Proposition 2.2 establishes that expansion decisions are interdependent across markets.

It is crucial to distinguish this congestion effect from the return to the blueprint itself. Because the blueprint is non-rival, expanding scope increases the marginal benefit of investing in blueprint quality. For example, consider a firm headquartered in the United States that initially operates in the EU and subsequently expands to China. Entering China increases the scale over which the blueprint can be deployed, strengthening the incentive to improve z . At the same time, serving China raises organizational complexity, tightening the firm's organizational constraint and increasing the cost of adaptation in all markets, including the EU. Proposition 2.2 captures this latter force holding z fixed, while Proposition 2.3 captures the countervailing effect: the return to blueprint investment rises with the firm's scope. Whether incumbent markets become more or less profitable in equilibrium depends on the balance between these two forces.

Proposition 2.3 (Returns to blueprint scale). *Fix an organizational complexity level $C \geq 0$. Suppose Assumption 2.4 holds. Then for any served set $M \subseteq S$, the marginal return to blueprint quality*

$$\frac{\partial}{\partial z} \int_{s \in M} \Pi_s(z; C) \mu(ds)$$

is weakly increasing in M with respect to set inclusion. In particular, if $M_1 \subseteq M_2$, then

$$\frac{\partial}{\partial z} \int_{s \in M_2} \Pi_s(z; C) \mu(ds) \geq \frac{\partial}{\partial z} \int_{s \in M_1} \Pi_s(z; C) \mu(ds).$$

Proof. See Appendix A.2. □

Because the blueprint is non-rival, a marginal improvement in blueprint quality raises effective productivity in every market where it is deployed. Serving a larger set of markets therefore increases the aggregate return to blueprint investment, holding organizational complexity fixed. In our earlier quantitative work (Dyrda, Hong, Sajid, and Steinberg, 2025), this mechanism appeared in a highly stylized and tightly parametrized environment with explicit production technologies, discrete multinational expansion, and general equilibrium feedbacks. In that setting, scalable complementarity between intangible investment and multinational expansion emerged from specific functional forms and calibration choices. Proposition 2.3 shows that this logic is considerably more general: it follows directly from non-rival deployment at the organizational level and does not rely on any particular market structure, production technology, or parametric specification.

Net effect of scope expansion. Proposition 2.2 shows that, holding blueprint quality fixed, expanding the firm’s footprint raises organizational complexity and thereby weakly lowers the value of all incumbent markets. Proposition 2.3 shows that expanding scope raises the marginal return to investing in non-rival blueprint quality. When blueprint quality is chosen optimally, these forces operate in opposite directions.

For any incumbent market $s \in M$, the change in its value following an expansion $M' = M \cup \Delta$ admits the decomposition

$$\Delta_{\Pi} = \underbrace{\left[\Pi_s((z^*(M'); C(M')) - \Pi_s((z^*(M); C(M')) \right]}_{\text{blueprint adjustment}} + \underbrace{\left[\Pi_s((z^*(M); C(M')) - \Pi_s((z^*(M); C(M)) \right]}_{\text{organizational congestion}}. \quad (13)$$

The second term is weakly negative by Proposition 2.2. The first term captures the effect of adjusting blueprint quality at the new complexity level and is weakly positive whenever scope expansion

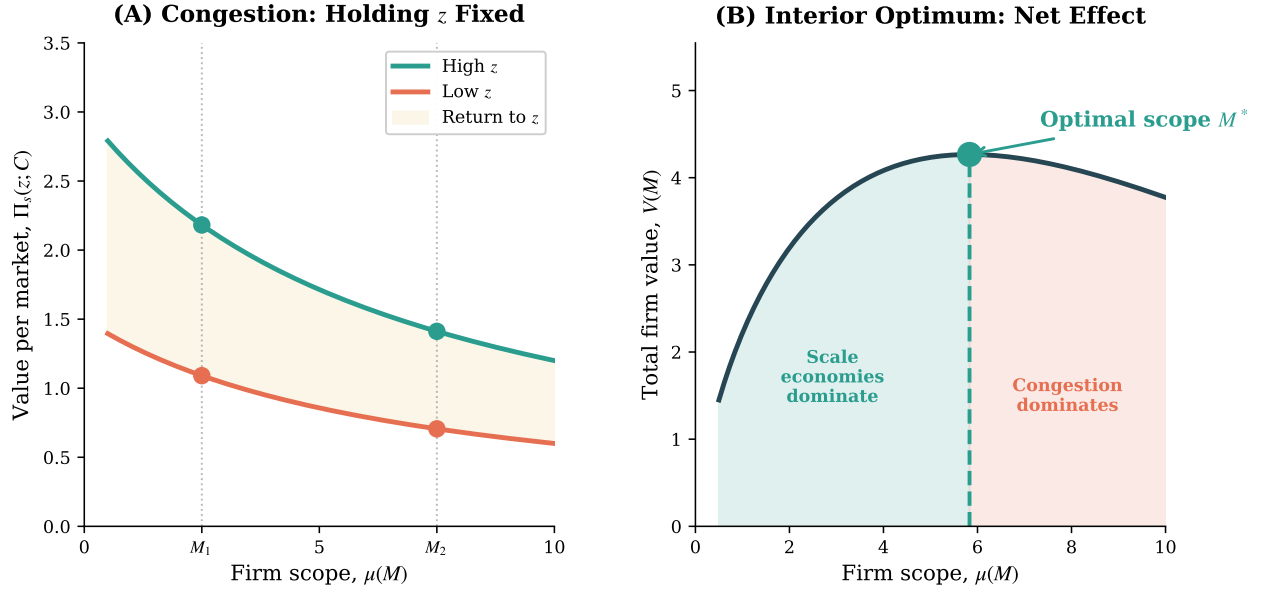


Figure 3: The scope–technology trade-off

induces an increase in the optimal blueprint. As a result, the net effect of scope expansion on incumbent market performance is a priori ambiguous and depends on the balance between these two forces.

Taken together, Propositions 2.2 and 2.3 characterize the opposing effects of scope expansion on incumbent market values and on the incentives to invest in organizational technology. Given these forces, the firm's footprint choice reduces to a selection problem over markets: which markets to serve, taking blueprint quality and organizational complexity as given. The next result shows that, under a mild ordering assumption on market heterogeneity, this selection problem admits a simple cutoff structure.

Assumption 2.6 (Heterogeneity ordering). *Markets can be ranked by heterogeneity $d(s)$. Fix (z, C) . The market-level value $\Pi_s(z; C)$ is weakly decreasing in $d(s)$: if $d(s_1) \leq d(s_2)$ then $\Pi_{s_1}(z; C) \geq \Pi_{s_2}(z; C)$.*

Assumption 2.6 ensures that markets can be ordered by organizational difficulty, so that more heterogeneous markets are weakly less profitable to serve for a given blueprint and organizational capacity. This assumption provides a monotone structure to the extensive margin of expansion without imposing any restrictions on returns to scale or within-market technologies.

Proposition 2.4 (Cutoff selection in heterogeneity). *Fix blueprint quality $z \geq 0$. Suppose Assumption 2.6 holds. Then the firm's optimal footprint M^* is monotone in heterogeneity: if $s_2 \in M^*$ and $d(s_1) \leq d(s_2)$, then $s_1 \in M^*$. Equivalently, there exists a threshold $d^* \in [0, \infty]$ such that*

$$s \in M^* \iff d(s) \leq d^*,$$

up to μ -null sets.

Proof. See Appendix A.2. □

Interpretation. Proposition 2.4 shows that optimal market selection is ordered by heterogeneity. Because organizational complexity increases with both the scope and the composition of activities, serving more heterogeneous markets imposes a disproportionately larger organizational burden on the firm. At the same time, holding blueprint quality fixed, the value of operating a market is lower when that market is more distant from the blueprint's origination environment. The firm therefore expands selectively, serving activities that are sufficiently similar to its core operations and excluding those that are too heterogeneous.

Importantly, this cutoff structure does not rely on fixed costs, decreasing returns, or independent market profitability. Instead, it arises from organizational scarcity: expansion tightens a shared internal constraint, and markets are selected based on their contribution to overall organizational complexity. As a result, firm growth follows an ordered pattern—near or closely related activities are served before distant or heterogeneous ones—even though the blueprint itself is non-rival.

This mechanism aligns closely with recent empirical evidence documenting that modern firms expand scope primarily by entering highly related product markets rather than by diversifying into unrelated activities. Using text-based measures of product market scope, [Hoberg and Phillips](#)

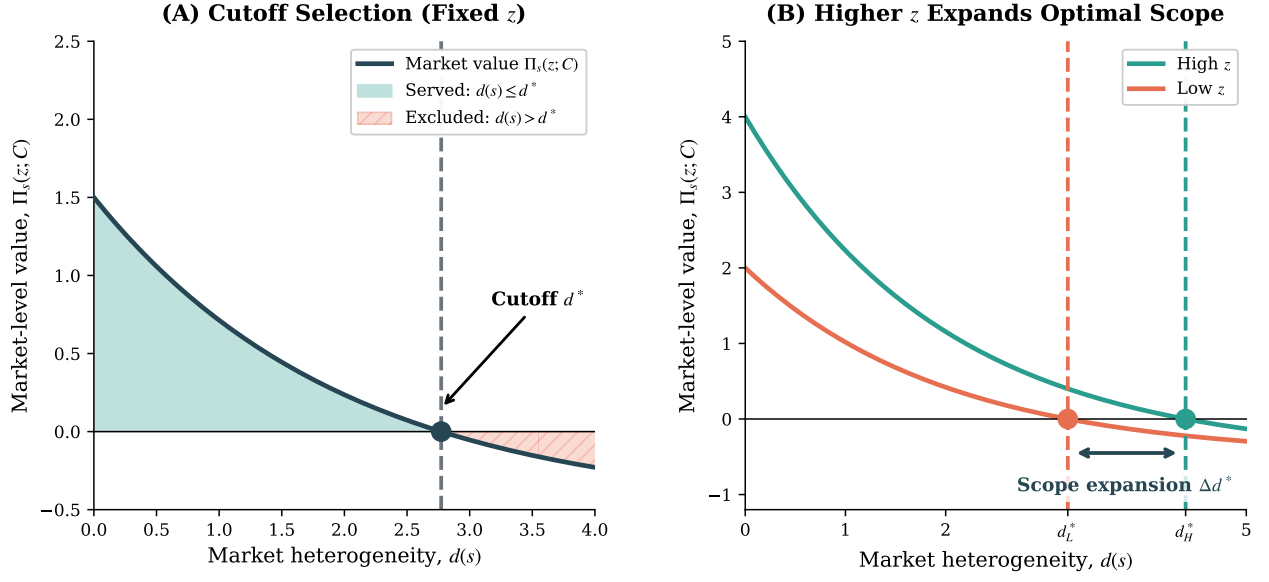


Figure 4: Cutoff selection in heterogeneity

(2025) show that scope expansion over the past several decades has been concentrated in related markets and has occurred largely without increases in traditional operating segments. Proposition 2.4 provides a theoretical foundation for this pattern: when organizational capacity is scarce, firms optimally prioritize related expansions that add less organizational complexity per unit of scope.

2.3 Adaptation and Core–Periphery Organization

Assumption 2.7 (Adaptation regularity). *The functions $R(z_s, x)$, $\theta(a, d)$, and $A(h, \ell)$ are continuously differentiable in their relevant arguments, with $R_1(z_s, x) \geq 0$, $\partial_a \theta(a, d) \geq 0$, and $\partial_h A(h, \ell) > 0$.*

Interpretation. Assumption 2.7 is a regularity condition ensuring that market-level payoffs and effective blueprint deployment respond smoothly to changes in headquarters attention and local adaptation effort. The assumption guarantees that organizational effort is productive at the margin and allows us to characterize how scarce headquarters attention is allocated across markets, without imposing curvature or restrictions on returns to scale.

Assumption 2.8 (Single-crossing in heterogeneity). *For any fixed $a \geq 0$, the marginal effectiveness of adaptation is weakly decreasing in heterogeneity: if $d_1 \leq d_2$, then*

$$\partial_a \theta(a, d_1) \geq \partial_a \theta(a, d_2).$$

Assumption 2.8 imposes a single-crossing property in market heterogeneity, so that headquarters-driven adaptation is relatively more effective in less heterogeneous markets. This structure delivers monotone allocation of organizational resources within the firm's footprint while remaining agnostic about within-market technologies.

Proposition 2.5 (Core–periphery allocation of headquarters attention). *Fix blueprint quality $z \geq 0$, organizational complexity C , and a served set M . Consider the market-level inner problems defining $\Pi_s(z; C)$ for $s \in M$.*

Suppose Assumptions 2.7–2.8 hold and that, for each $s \in M$, the optimal choice of headquarters attention $h^(s)$ is interior. Then headquarters attention allocated across markets is weakly decreasing in heterogeneity: for any $s_1, s_2 \in M$,*

$$d(s_1) \leq d(s_2) \implies h^*(s_1) \geq h^*(s_2).$$

Proof. See Appendix A.3. □

Proposition 2.5 characterizes how a firm allocates scarce organizational resources within its optimal footprint. Because headquarters attention is a shared and costly input, the firm cannot adapt all markets equally. Markets that are closer to the blueprint's origination environment have higher marginal returns to headquarters-driven adaptation, as organizational effort translates more effectively into usable blueprint deployment. By contrast, in more heterogeneous markets, the same organizational effort yields smaller gains. The firm therefore concentrates headquarters attention in low-heterogeneity core markets and operates high-heterogeneity peripheral markets with less intensive organizational involvement.

This core–periphery structure does not arise from fixed costs, technological constraints, or differences in market size. Rather, it is an endogenous response to organizational scarcity: as the firm expands and organizational complexity rises, headquarters attention is rationed toward activities where it is most productive. As a result, firms appear centralized in their core operations while maintaining broader, more standardized peripheral activities. This pattern is consistent with empirical evidence showing that large firms tend to concentrate managerial oversight and customization in core activities while relying on standardized, less intensive organizational processes in more peripheral lines of business.

Corollary 2.2 (Effectiveness gradient). *Under the conditions of Proposition 2.5, effective blueprint*

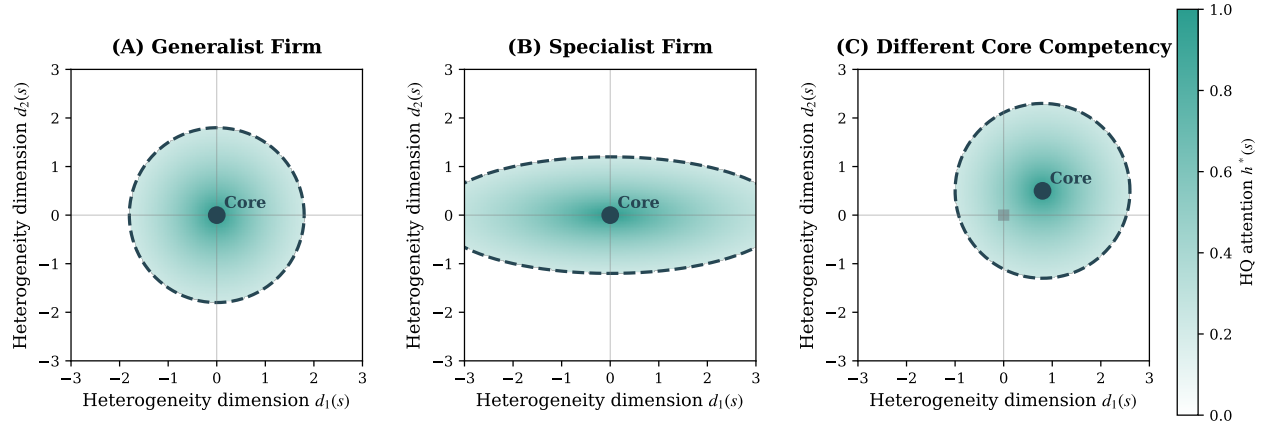


Figure 5: Core-periphery organization within the firm

deployment is weakly decreasing in heterogeneity:

$$d(s_1) \leq d(s_2) \implies z_{s_1}^* \geq z_{s_2}^*.$$

The core-periphery allocation of headquarters attention implies an effectiveness gradient across markets. Activities that are closer to the firm's organizational core operate with higher effective blueprint deployment, while more heterogeneous peripheral activities are served in a more standardized and less effective manner. Organizational complexity therefore shapes not only which markets are served, but also how intensively and effectively they are operated.

Remark 2.9 (Standardization as a substitute for managerial attention). *The production function for effective deployment, $z_s = \theta(A(h(s), \ell(s)), d(s)) z$, implies a substitution between blueprint quality z and headquarters attention $h(s)$. To achieve a given level of effective deployment in a heterogeneous market, the firm can either allocate rival organizational effort to adapt the blueprint (raising the portability term θ) or invest in the non-rival quality of the blueprint itself (raising z).*

As a result, improvements in organizational technology reduce the marginal need for bespoke managerial adaptation in heterogeneous markets. This substitution underlies the core-periphery allocation of headquarters attention and foreshadows how investments in non-rival organizational capital relax internal capacity constraints.

3 Platform Emergence

The results in Section 2 characterize firm scope and internal organization when organizational complexity imposes a binding constraint on expansion. Even though firms may operate across many highly similar activities, the aggregate heterogeneity of the footprint must remain finite, and organizational scarcity ultimately limits scope. In this baseline environment, firms differ in how broadly they expand, but all face the same fundamental trade-off between scope and organizational congestion.

This section asks whether that trade-off can be endogenously relaxed. We introduce an organizational technology that allows firms to reduce the effective heterogeneity associated with serving markets. Unlike the adaptation choices studied in Section 3, which allocate scarce headquarters attention across activities, organizational technology operates at a distinct margin: it reshapes how market heterogeneity aggregates into organizational complexity. As a result, such investments do not merely lower the level of complexity; they can alter how complexity scales with scope.

The central question is whether sufficiently strong organizational technology can overturn the organizational constraint that governs scope in the baseline model. We show that, under mild conditions, this is possible. Beyond a critical threshold, firms can expand their footprint without incurring proportionate increases in organizational complexity, generating a qualitative transition from bounded to unbounded scope. We interpret this transition as the emergence of platform-type organizations: firms whose internal structure allows them to scale across arbitrarily many markets while maintaining finite organizational burden.

Importantly, platform emergence in this framework does not reflect a smooth decline in marginal costs. Instead, it arises from the endogenous collapse of an internal organizational constraint. The remainder of this section formalizes this mechanism and characterizes the conditions under which organizational technology gives rise to unbounded scope.

3.1 Extended model: endogenous heterogeneity reduction

We extend the baseline environment by allowing firms to invest in standardization that reduces the effective heterogeneity of served markets.

Effective heterogeneity and deployment. Given market-specific standardization intensity $\sigma(s) \geq 0$, define effective heterogeneity

$$\tilde{d}(s) \equiv d_0(s) g(\sigma(s)).$$

Effective blueprint deployment in market s is then

$$z_s = \theta(A(h(s), \ell(s)), \tilde{d}(s)) z = \theta(A(h(s), \ell(s)), d_0(s) g(\sigma(s))) z. \quad (14)$$

Standardization costs. Standardization is costly and interacts with blueprint quality. We assume separability

$$\kappa(\sigma, z) = \psi(z) c(\sigma),$$

where c is increasing and convex with $c(0) = 0$, and ψ is decreasing with $\lim_{z \rightarrow \infty} \psi(z) = 0$. Intuitively, higher blueprint quality reduces the unit cost of standardizing additional markets.

Complexity with standardization. Organizational complexity of a served set M is now

$$C(M, \sigma) \equiv \int_{s \in M} \tilde{d}(s) ds = \int_{s \in M} d_0(s) g(\sigma(s)) ds.$$

Remark 3.1 (Relation to standardization-based scale models). *Recent work, most notably [Argente et al. \(2025\)](#), emphasizes standardization as a force that lowers marginal costs and facilitates firm expansion by flattening cost schedules as production becomes more replicable.*

Our framework also features organizational costs, captured by the complexity-dependent shadow price $q(C)$ of headquarters attention, but standardization operates through a different channel. Rather than reducing marginal costs directly, standardization lowers effective heterogeneity, which governs how organizational complexity aggregates across activities. When low-heterogeneity opportunities are sufficiently dense, firms can expand scope along margins that add little to organizational burden, even though headquarters attention remains scarce. This mechanism relaxes internal organizational constraints they scale with scope, rather than by flattening production costs.

Firm problem. The firm chooses (z, M) and measurable allocations $\{x(s), \ell(s), h(s), \sigma(s)\}_{s \in M}$ to maximize

$$\int_{s \in M} R(z_s, x(s)) ds - q(C(M, \sigma)) \left[\chi(z) + \int_{s \in M} h(s) ds \right] - \int_{s \in M} \kappa(\sigma(s), z) ds,$$

subject to the baseline deployment and feasibility constraints. In the extended model, the market heterogeneity argument in the portability function is the effective heterogeneity $\tilde{d}(s)$, so that $z_s = \theta(A(h(s), \ell(s)), \tilde{d}(s)) z$.

3.2 Resulte on platform emergence

Assumption 3.2 (Density of low-heterogeneity opportunities). *For every $\varepsilon > 0$, the sublevel set*

$$S_\varepsilon \equiv \{s \in S : d_0(s) < \varepsilon\}$$

has infinite measure: $\mu(S_\varepsilon) = \infty$.

Assumption 3.2 requires that markets arbitrarily similar to the firm’s organizational core exist in unbounded measure. It captures environments with rich near-core granularity: product variants, customer segments, geographic subdivisions, or standardized service requests that differ only along minor dimensions. This is a property of the market space, not of firm behavior or profitability. Absent this density property—for instance, if baseline heterogeneity were bounded away from zero, $d_0(s) \geq \underline{d} > 0$ for all s —platform-type expansion would be impossible regardless of organizational technology, because each additional market would impose a strictly positive irreducible complexity cost.

Assumption 3.3 (Near-core deployment efficiency). *The deployment technology satisfies:*

- (i) **Perfect portability at zero heterogeneity:** $\theta(a, 0) = 1$ for all $a > 0$.
- (ii) **Continuity and monotonicity:** For each $a > 0$, $\theta(a, d)$ is continuous and nonincreasing in d .
- (iii) **Uniform feasible recipe:** There exist $(\bar{x}, \bar{h}, \bar{\ell})$ with $\bar{x}, \bar{h} > 0$, $\bar{\ell} \geq 0$ such that

$$\bar{x} + \bar{h} + \bar{\ell} \leq \inf_{s \in S} \bar{X}(s), \quad A(\bar{h}, \bar{\ell}) > 0.$$

- (iv) **Zero standardization cost at zero:** $c(0) = 0$.

Interpretation. Assumption 3.3 imposes regularity conditions on the deployment technology that ensure near-core markets can be served using a common, well-behaved organizational recipe. Part (i) normalizes deployment in markets identical to headquarters, defining the operational meaning of zero heterogeneity. Part (ii) ensures that deployment efficiency deteriorates continuously—and weakly monotonically—as market heterogeneity increases, so sufficiently similar markets admit uniformly high portability. Part (iii) guarantees the existence of a fixed, feasible production and adaptation plan that can be implemented uniformly across markets and delivers strictly positive adaptation. Part (iv) ensures that abstaining from standardization entails no direct cost. Together, these conditions ensure that near-core markets can be served in a uniform manner without

Assumption 3.4 (Increasing returns to blueprint quality). *There exists $\bar{x} > 0$ such that:*

(i) **Unbounded revenues.**

$$\lim_{z \rightarrow \infty} R(z, \bar{x}) = \infty.$$

(ii) **Revenue growth dominates blueprint creation cost.** *For any $\alpha \in (0, 1]$*

$$\lim_{z \rightarrow \infty} \frac{R(\alpha z, \bar{x})}{\chi(z)} = \infty.$$

Interpretation. Assumption 3.4 contains the sole source of increasing returns in the model and is the key economic condition underlying platform feasibility. Part (i) requires that improvements in blueprint quality raise revenues without bound for a fixed production scale. Part (ii) strengthens this by requiring that revenue growth eventually dominates the fixed cost of blueprint creation. Together, these conditions formalize increasing returns to organizational technology: the blueprint is a non-rival input whose benefits scale with deployment, while creation costs are incurred only once.

While Assumption 3.4 strengthens the standard increasing returns condition, it imposes no additional restrictions on the iso-elastic functional forms typically employed in quantitative macroeconomics. For standard power functions—where $R(z, \bar{x}) \propto z^\gamma$ and $\chi(z) \propto z^\beta$ with $\gamma > \beta$ —the condition is automatically satisfied for any $\alpha > 0$, because $R(\alpha z) \propto \alpha^\gamma z^\gamma$. In this case, the portability friction α acts merely as a level shifter and does not alter the asymptotic dominance of revenues over creation costs. The assumption becomes restrictive only for functional forms where deployment frictions dampen the *rate* of growth rather than its level, such as exponential specifications (e.g., $R(z) \propto e^{\gamma z}$) where $\alpha < 1$ would reduce the effective growth rate below that of costs. Consequently, the assumption serves primarily to ensure that the *curvature* of returns to scale sufficiently exceeds the curvature of creation costs, ensuring that the increasing returns mechanism is robust to the inevitable efficiency losses inherent in serving heterogeneous peripheral markets.

Lemma 3.1 (Uniform near-core surplus). *Suppose Assumptions 3.3 and 3.4 hold. Then for any $\bar{C} < \infty$, there exist constants $\varepsilon > 0$, $z_1 < \infty$, and a function $\underline{\pi} : [z_1, \infty) \rightarrow (0, \infty)$ such that:*

(i) *For all $z \geq z_1$, all $C \in [0, \bar{C}]$, and all markets s with $d_0(s) \leq \varepsilon$,*

$$\hat{\Pi}_s(z; C) \geq \underline{\pi}(z) > 0.$$

(ii) The function $\underline{\pi}(z)$ satisfies

$$\lim_{z \rightarrow \infty} \underline{\pi}(z) = \infty \quad \text{and} \quad \lim_{z \rightarrow \infty} \frac{\underline{\pi}(z)}{\chi(z)} = \infty.$$

Proof. See Appendix A.4. □

Lemma 3.1 shows that sufficiently similar markets can be served using a common organizational recipe that delivers a uniformly positive surplus once blueprint quality is high enough. The surplus bound is uniform across all near-core markets and robust to any bounded level of organizational congestion. Moreover, the per-market surplus grows without bound with blueprint quality and eventually dominates the fixed cost of blueprint creation. This result isolates the economic mechanism underlying platform feasibility: increasing returns to blueprint quality translate into scalable, market-by-market profitability in environments with abundant near-core opportunities.

Proposition 3.1 (Persistence under weak organizational technology). *Let $\liminf_{z \downarrow 0} \psi(z) > 0$. Then there exists $\underline{z} > 0$ such that for every $z \in (0, \underline{z}]$, any optimal solution $(M^*(z), \sigma^*(\cdot), x^*(\cdot), \ell^*(\cdot), h^*(\cdot))$ of the firm's problem in the extended environment satisfies:*

- (i) **No standardization:** $\sigma^*(s) = 0$ for μ -a.e. $s \in M^*(z)$, hence $\tilde{d}(s) = d_0(s)$ for μ -a.e. $s \in M^*(z)$.
- (ii) **Reduction to the baseline problem:** the induced market-level payoffs coincide with those in the baseline environment of Section 2.
- (iii) **Bounded organizational complexity:** the optimal complexity is finite,

$$C(M^*(z), \sigma^*) = \int_{M^*(z)} d_0(s) g(\sigma^*(s)) ds = \int_{M^*(z)} d_0(s) ds < \infty.$$

Proof. See Appendix A.4. □

Proposition 3.1 establishes that allowing for standardization does not, by itself, relax the organizational constraint identified in Section 2. When blueprint quality is low, standardization remains prohibitively costly, so the firm optimally refrains from using it and the extended model collapses to the baseline environment. As a result, the bounded-complexity conclusion of Proposition 2.1 applies unchanged: organizational congestion continues to bind and prevents unbounded expansion. This result clarifies that platform-type behavior is not an automatic consequence of adding organizational instruments; escaping the complexity constraint requires sufficiently strong organizational technology, which is characterized in the subsequent propositions.

Proposition 3.2 (Platform feasibility). *Suppose Assumptions 3.2, 3.3, and 3.4 hold, and let Lemma 3.1 apply. Then for any target scale $K < \infty$, there exist $z_K < \infty$ and $\bar{C} < \infty$ such that for all $z \geq z_K$ the firm can choose a measurable set $M \subseteq S$ and allocations (x, ℓ, h, σ) satisfying*

$$\mu(M) \geq K, \quad C(M, \sigma) = \int_M d_0(s)g(\sigma(s)) ds \leq \bar{C},$$

and achieving strictly positive total payoff (net of congestion and standardization costs). In particular, for all $z \geq z_K$ there exists a feasible plan with footprint at least K and finite organizational complexity.

Proof. See Appendix A.4. □

Proposition 3.2 shows that sufficiently strong organizational technology makes large-scale operation feasible despite organizational congestion. When blueprint quality is high enough, the firm can profitably replicate a common operating recipe across an arbitrarily large collection of near-core markets while keeping organizational complexity finite. The result isolates the conditions under which scale becomes technologically feasible: abundant near-core opportunities ensure that expansion does not mechanically raise complexity, while increasing returns to blueprint quality ensure that the surplus generated in each market dominates the fixed cost of blueprint creation. Importantly, the proposition is purely about feasibility rather than optimality—it establishes that platform-scale operation is possible, not that it is necessarily chosen—setting the stage for the emergence result in Proposition 3.3.

Proposition 3.3 (Platform emergence threshold). *Suppose Assumptions 3.2, 3.3, and 3.4 hold, and let $\liminf_{z \downarrow 0} \psi(z) > 0$. Then there exists a threshold $z^* \in (0, \infty)$ such that:*

- (i) **Bounded regime.** *For all $z \in (0, z^*)$, every optimal solution has bounded footprint, $\mu(M^*(z)) < \infty$.*
- (ii) **Platform regime.** *For all $z > z^*$, there exists an optimal solution that exhibits platform behavior in the sense that for every $K < \infty$ the firm can achieve $\mu(M) \geq K$ with finite organizational complexity and strictly higher value than any bounded-footprint alternative.*

Proof. See Appendix A.4. □

Proposition 3.3 characterizes platform emergence as a phase transition driven by organizational technology. When blueprint quality is low, standardization is ineffective and the firm remains subject to the organizational complexity constraint identified in Section 2, resulting in bounded scope.

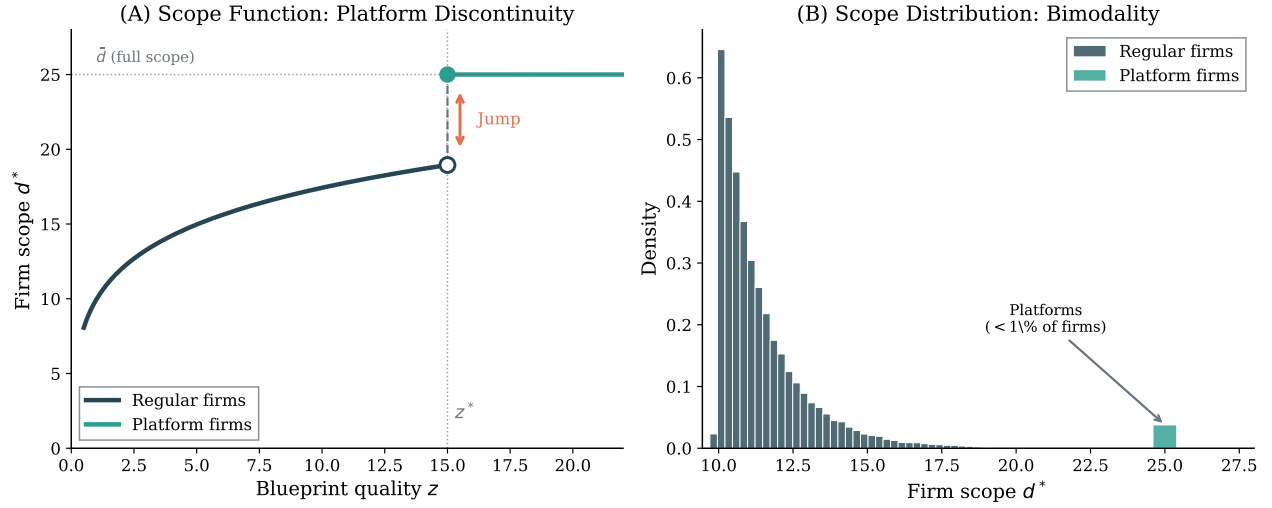


Figure 6: Organizational complexity compresses the firm scope distribution.

Notes:

As blueprint quality improves, increasing returns to organizational technology and the availability of abundant near-core markets make large-scale replication feasible. Beyond the threshold z^* , the marginal gains from expanding into additional near-core markets remain positive despite congestion, so platform-type expansion strictly dominates any bounded alternative. The proposition formalizes the idea that platforms are not merely large firms, but firms that have crossed a technological threshold at which organizational constraints cease to bind at the margin.

Throughout this section, blueprint quality z is treated as an exogenous technology parameter. Proposition 3.3 therefore characterizes how optimal firm scope varies across firms or environments with different levels of organizational technology.

4 Implications for the Firm Scope Distribution

This section characterizes how the organizational complexity mechanism shapes the cross-sectional distribution of firm scope. We show that the model generates a distinct distributional signature—compressed right tails—that differs qualitatively from standard scale models and provides a structural explanation for the observed scarcity of very large firms.

4.1 From Blueprint Quality to Scope

We begin by characterizing the mapping from blueprint quality to optimal firm scope. Consider a population of firms indexed by blueprint quality z , drawn from a distribution F_z with support

$[z, \infty)$. Each firm solves the problem in Section 1, yielding an optimal footprint $M^*(z)$ and associated scope $\mu(M^*(z))$.

By Proposition 2.4, the optimal footprint takes a cutoff form in the heterogeneity index: $M^*(z) = \{s \in S : d(s) \leq d^*(z)\}$ for some threshold $d^*(z)$ that depends on blueprint quality. We can therefore define the scope function directly in terms of this cutoff.

Definition 3 (Scope function). *The scope function $\Phi : [z, \infty) \rightarrow \mathbb{R}_+$ maps blueprint quality to optimal firm scope:*

$$\Phi(z) \equiv \mu(M^*(z)) = \mu(\{s \in S : d(s) \leq d^*(z)\}), \quad (15)$$

where $d^*(z)$ is the optimal heterogeneity cutoff from Proposition 2.4.

The scope function inherits its shape from two primitives: (i) how the optimal cutoff $d^*(z)$ responds to blueprint quality, and (ii) the distribution of markets by heterogeneity.

To fix ideas, consider the case where markets are indexed directly by their heterogeneity, $s \in [0, \bar{d}]$ with $d(s) = s$, and have uniform density. Then $\Phi(z) = d^*(z)$, and the scope function is determined entirely by the first-order condition for the optimal cutoff.

Contrast with standard models. In canonical models of firm size based on span-of-control (Lucas, 1978) or monopolistic competition (?), the mapping from productivity to size is typically log-linear: if firm size is z^γ for some $\gamma > 0$, then a Pareto distribution of productivity translates directly into a Pareto distribution of size, with tail index α_z/γ . The tail may be fatter or thinner depending on γ , but the Pareto form is preserved. There is no natural “speed limit” on how differences in productivity translate into differences in size—a firm with $10\times$ the productivity of another will be 10^γ times as large.

The organizational complexity mechanism introduces such a speed limit. The constraint is not technological (decreasing returns to scale) but organizational: managing a diverse portfolio of heterogeneous markets becomes progressively harder as the firm expands into more distant territory. Crucially, this difficulty arises from market heterogeneity—the fact that the marginal market for a large firm is more distant from its organizational core than the marginal market for a small firm.

4.2 The Scope Elasticity

The key object governing the transformation from productivity to scope is the scope elasticity.

Definition 4 (Scope elasticity). *The scope elasticity at blueprint quality z is*

$$\varepsilon_\Phi(z) \equiv \frac{z \Phi'(z)}{\Phi(z)} = \frac{d \log \Phi(z)}{d \log z}. \quad (16)$$

The scope elasticity measures the percent increase in scope from a one percent increase in blueprint quality. In standard models with $\Phi(z) = z^\gamma$, the elasticity is constant: $\varepsilon_\Phi(z) = \gamma$ for all z . The following result shows that heterogeneity in markets generates a declining elasticity.

Proposition 4.1 (Declining scope elasticity). *Suppose markets are heterogeneous with $d(s)$ distributed over $[0, \bar{d}]$, and that:*

- (i) *Expansion is ordered by heterogeneity: $d^*(z)$ is increasing in z .*
- (ii) *Portability decays with distance: $\theta'(d) < 0$ for all $d \in [0, \bar{d}]$.*
- (iii) *Organizational congestion is increasing: $q'(C) > 0$ for all $C \geq 0$.*

Then the scope elasticity $\varepsilon_\Phi(z)$ is strictly decreasing in z for $z \in [\underline{z}, z^)$.*

The declining elasticity arises entirely from market heterogeneity and the ordered structure of expansion. No curvature assumptions on the congestion function $q(\cdot)$ or technology are required beyond $q' > 0$. The mechanism is that high- z firms have already served the “easy” (low- d) markets; further expansion requires venturing into “harder” (high- d) territory where portability is lower and complexity costs are higher. This is a pure composition effect: the quality of the marginal market deteriorates as scope expands.

4.3 Tail Compression

The declining scope elasticity has direct implications for the distribution of firm scope.

Proposition 4.2 (Tail compression). *Suppose blueprint quality z is drawn from a Pareto distribution with tail index $\alpha_z > 0$:*

$$\Pr(z > x) = \left(\frac{\bar{z}}{x}\right)^{\alpha_z} \quad \text{for } x \geq \bar{z}. \quad (17)$$

Under the conditions of Proposition 4.1, the distribution of firm scope $\Phi(z)$ has a strictly thinner right tail than Pareto with index α_z . Specifically:

- (i) *The tail index of the scope distribution satisfies $\alpha_M > \alpha_z$.*
- (ii) *For any $\varepsilon > 0$, $\Pr(\Phi(z) > x) = o(x^{-\alpha_z + \varepsilon})$ as $x \rightarrow \bar{d}$.*
- (iii) *If $\bar{d} < \infty$, the scope distribution has bounded support.*

Proposition 4.2 delivers the paper’s central distributional result: organizational complexity compresses the right tail of the firm scope distribution. Even if blueprint quality has a fat-tailed Pareto distribution—as commonly assumed and empirically supported—the distribution of firm scope will have a thinner tail.

The intuition is that high- z firms face diminishing returns to expansion not because of technology, but because the marginal market gets harder as the firm grows. A firm with $z = 100$ has already served all low-heterogeneity markets; its marginal expansion is into high- d territory where portability is low and complexity costs are high. A firm with $z = 10$ faces a much more favorable marginal market. The same proportional increase in z therefore yields a smaller proportional increase in scope for the larger firm.

This mechanism explains why we observe fewer very large firms than a Pareto productivity distribution would predict—the “missing giants” in the right tail of the firm size distribution.

4.4 Illustration

We illustrate the tail compression mechanism using a parametric specification. Suppose markets are indexed by heterogeneity $s \in [0, \bar{d}]$ with $d(s) = s$, portability takes the exponential form $\theta(s) = e^{-s/\lambda}$, and organizational complexity is $C(d^*) = (d^*)^2/2$. The congestion function is $q(C) = (1 + \kappa C)^\phi$, where $\kappa > 0$ governs the intensity of organizational congestion. Each market requires fixed headquarters attention $\bar{h}$, so the marginal cost of serving a market at distance d^* is $q(C(d^*))\bar{h}$ plus the indirect cost from raising complexity throughout the firm.

Figure 7 illustrates the central result. Panel (A) plots the scope function $\Phi(z) = d^*(z)$. Two firms are marked: Firm A with blueprint quality $z = 2$ achieves optimal scope $d^* = 12$, while Firm B with $z = 10$ achieves $d^* = 17$. The key observation is that Firm B has five times the blueprint quality but only 1.5 times the scope. This compression arises because organizational constraints bind more tightly as firms expand: portability decays in distant markets, complexity accumulates, and congestion raises costs throughout the organization. The scope function is concave—better firms expand more, but at a diminishing rate.

Panel (B) shows the distributional consequence. We draw blueprint quality z from a Pareto distribution with tail index $\alpha_z = 2$ and compute optimal scope for each firm. The panel plots complementary CDFs on a log-log scale, where Pareto distributions appear as straight lines with slope equal to the negative tail index. The input distribution (blueprint quality, dark line) has slope -2 . The output distribution (firm scope, teal line) has slope approximately -9 —more than four times steeper. A steeper slope means faster tail decay: at any threshold, far fewer firms exceed that

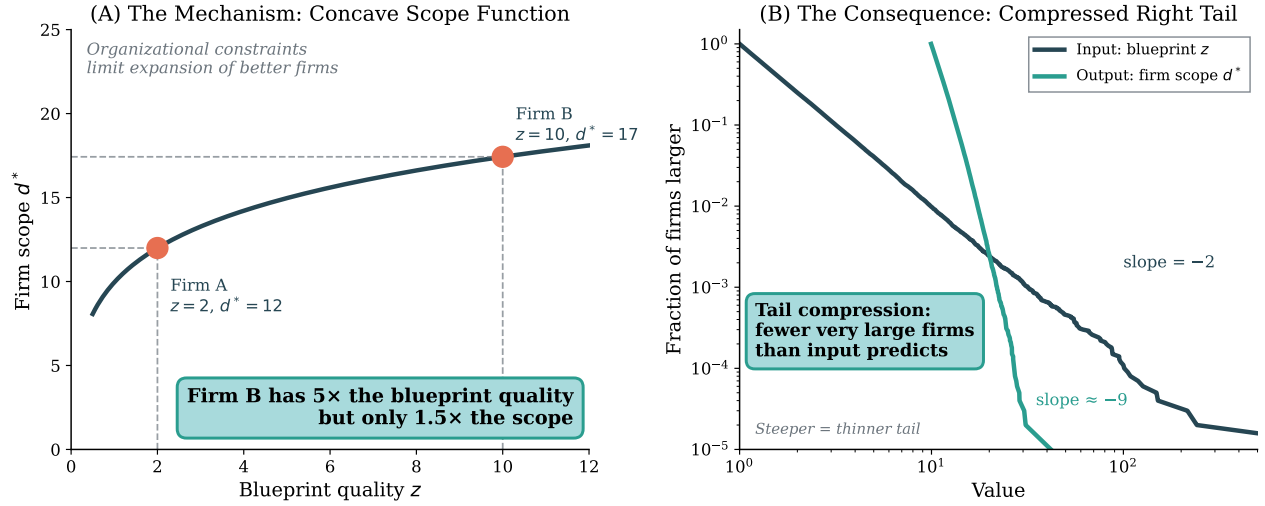


Figure 7: Organizational complexity compresses the firm scope distribution

Panel (A): The scope function $\Phi(z)$ is concave. Firm B has 5× the blueprint quality of Firm A but only 1.5× the scope, because organizational constraints bind more tightly for larger firms. Panel (B): Complementary CDFs on log-log scale. The input (blueprint quality z) follows a Pareto distribution with tail index $\alpha_z = 2$. The output (firm scope d^*) has a much steeper slope (≈ -9), indicating a thinner tail—fewer very large firms than the input distribution would predict.

threshold in scope than would be predicted by applying the input distribution directly. This is the “missing tail” generated by organizational complexity.

The magnitude of compression depends on model parameters. Stronger organizational congestion (higher κ) flattens the scope function further, producing even thinner tails. This suggests a testable cross-sectional prediction: industries or environments with more severe organizational frictions should exhibit thinner-tailed scope distributions, holding the distribution of firm capabilities fixed. Conversely, industries where coordination is easier—perhaps due to modular production or standardized interfaces—should display fatter tails and more dispersion in firm scope.

5 Conclusion

This paper has proposed organizational complexity as the binding constraint on firm scope. The central insight is that the cost of expansion depends not on how many markets a firm serves, but on the cumulative heterogeneity of those markets relative to the firm’s organizational core. This shift—from count to complexity—resolves a puzzle that count-based theories cannot address: why firms can scale across vast numbers of similar activities at negligible marginal cost, yet face sharp limits when expanding into materially distinct environments. The framework delivers a unified account of bounded scope, related diversification, core–periphery internal structure, and the discontinuous emergence of platform-type organizations.

The organizational complexity mechanism also reshapes how we think about the firm size distribution. In standard models, a fat-tailed distribution of productivity translates directly into a fat-tailed distribution of firm size. Here, that mapping is mediated by organizational constraints: larger firms face more heterogeneous marginal markets, so capability differences translate into progressively smaller scope differences as firms grow. The result is tail compression—fewer very large firms than the productivity distribution alone would predict. The “missing giants” in the right tail are not absent because highly capable firms do not exist, but because organizational complexity prevents capability from fully expressing itself as scope.

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Appendix

A Proofs

A.1 Proofs for Section 2.1.

Proof of Proposition 2.1. We proceed by contradiction. Suppose there exists an optimal solution $(M^*, z^*, \{x^*, \ell^*, h^*\}_{s \in M^*})$ with

$$C(M^*) = \int_{M^*} d(s) \mu(ds) = \infty.$$

Step 1: Total gross payoff is finite at the candidate optimum. Let $(M^*, z^*, \{x^*, \ell^*, h^*\})$ be the candidate optimal solution. By Assumption 1.7 (Conditional Integrability), for the fixed blueprint quality z^* , there exists an integrable envelope $\bar{R}_{z^*} \in L^1(S, \mu)$. Therefore, the total gross payoff over the served set M^* is finite at the candidate optimum:

$$\int_{M^*} R(z_s^*, x^*(s)) \mu(ds) \leq \int_{M^*} \bar{R}_{z^*}(s) \mu(ds) \equiv B(z^*) < \infty.$$

Note that while $B(z^*)$ is finite for the fixed z^* , we do not assume it is uniformly bounded for all z .

Step 2: Positive payoff in any $d(s) > 0$ market requires $h(s) > 0$. Fix a market s with $d(s) > 0$. By Assumption 2.2,

$$\theta(A(0, \ell), d(s)) = 0 \quad \forall \ell \geq 0.$$

Therefore, if $h^*(s) = 0$, then $a^*(s) = A(0, \ell^*(s))$ and the deployment constraint implies

$$z_s^* = \theta(a^*(s), d(s)) z^* = \theta(A(0, \ell^*(s)), d(s)) z^* = 0.$$

Since $x^*(s)$ is feasible (by (??)), Assumption 2.3 yields

$$R(z_s^*, x^*(s)) = R(0, x^*(s)) \leq 0.$$

Thus, for any market with $d(s) > 0$,

$$R(z_s^*, x^*(s)) > 0 \implies h^*(s) > 0. \tag{18}$$

Step 3: Any nontrivial optimum uses strictly positive total HQ attention. Let total HQ attention at (M^*, z^*, h^*) be

$$H^* \equiv \chi(z^*) + \int_{M^*} h^*(s) \mu(ds) \geq 0.$$

If $H^* = 0$, then $\chi(z^*) = 0$ implies $z^* = 0$ (since χ is increasing with $\chi(0) = 0$), and $h^*(s) = 0$ for μ -a.e. $s \in M^*$, implying $z_s^* = 0$ for μ -a.e. $s \in M^*$. Hence, by Assumption 2.3,

$$\int_{M^*} R(z_s^*, x^*(s)) \mu(ds) = \int_{M^*} R(0, x^*(s)) \mu(ds) \leq 0.$$

But the firm can always choose $M = \emptyset$ and $z = 0$, achieving payoff 0 at zero cost. Therefore, any optimal solution that attains a strictly positive objective value must satisfy $H^* > 0$.

Step 4: Infinite complexity implies arbitrarily large unit HQ cost. By assumption, $q'(C) > 0$ and $\lim_{C \rightarrow \infty} q(C) = \infty$. Since $C(M^*) = \infty$, we have

$$q(C(M^*)) = \infty,$$

in the sense that for every $B > 0$ there exists a (finite) complexity level $\tilde{C}$ large enough such that $q(\tilde{C}) > B$, and $C(M^*)$ exceeds $\tilde{C}$.

Step 5: Contradiction. The objective value at the candidate optimum equals

$$\int_{M^*} R(z_s^*, x^*(s)) \mu(ds) - q(C(M^*)) \left(\chi(z^*) + \int_{M^*} h^*(s) \mu(ds) \right) = \int_{M^*} R(z_s^*, x^*(s)) \mu(ds) - q(C(M^*)) H^*.$$

By Step 1, the first term is bounded above by $B(z^*) < \infty$. By Step 3, $H^* > 0$ for any nontrivial optimum. Since $q(C) \rightarrow \infty$ as $C \rightarrow \infty$ (Assumption 1.6) and $C(M^*) = \infty$, for any constant $\bar{B} > 0$ there exists a finite $\tilde{C}$ such that $q(\tilde{C}) > \bar{B}$, and $C(M^*) > \tilde{C}$. In particular, choosing $\bar{B} > B(z^*)/H^*$ yields

$$q(C(M^*)) H^* > B(z^*),$$

so that the objective value is strictly negative. This contradicts feasibility of $(M, z) = (\emptyset, 0)$, which yields value zero. Therefore $C(M^*) < \infty$ at any optimum.

□

Proof of Corollary 2.1. By definition of M_τ^* , we have $d(s) \geq \tau$ for all $s \in M_\tau^*$. Therefore

$$\int_{M^*} d(s) \mu(ds) \geq \int_{M_\tau^*} d(s) \mu(ds) \geq \int_{M_\tau^*} \tau \mu(ds) = \tau \mu(M_\tau^*).$$

Rearranging gives

$$\mu(M_\tau^*) \leq \frac{1}{\tau} \int_{M^*} d(s) \mu(ds).$$

By Proposition 2.1, $C(M^*) = \int_{M^*} d(s) \mu(ds) < \infty$, hence $\mu(M_\tau^*) < \infty$. $\square$

Remark A.1 (Complexity versus market count). *A common way to ensure finite scope in continuum environments is to impose a uniform lower bound on market heterogeneity, $d(s) \geq \underline{d} > 0$ μ -a.e. Under such an assumption, total organizational complexity and market measure coincide up to scale:*

$$C(M) = \int_M d(s) \mu(ds) \geq \underline{d} \mu(M),$$

so that $C(M) < \infty$ if and only if $\mu(M) < \infty$. In this case, bounding organizational complexity is equivalent to bounding the number (measure) of markets served, and the firm faces a hard span-of-control constraint whereby each additional market entails a strictly positive irreducible burden.

By contrast, when heterogeneity is allowed to vary across markets and may become arbitrarily small—while remaining integrable—finite complexity does not restrict the measure of markets served. Specifically, if $d \in L^1(S, \mu)$ and $d(s) > 0$ μ -a.e., then $C(M) < \infty$ is compatible with $\mu(M) = \infty$. Nevertheless, finite complexity still imposes sharp discipline on the number of materially distinct markets. For any threshold $\tau > 0$,

$$C(M) \geq \int_{\{s \in M : d(s) \geq \tau\}} d(s) \mu(ds) \geq \tau \cdot \mu(\{s \in M : d(s) \geq \tau\}),$$

which implies

$$\mu(\{s \in M : d(s) \geq \tau\}) \leq \frac{C(M)}{\tau}.$$

Thus, while the firm may serve a large (even unbounded) collection of markets with very small heterogeneity, it can serve only a finite mass of markets whose heterogeneity exceeds any fixed threshold τ . This distinction allows the model to accommodate scalable, platform-type organizations that operate across many highly similar markets, while preserving a meaningful notion of bounded organizational scope.

A.2 Proofs for Section 2.2

In proving Proposition 2.2 we will make use of the following standard result.

Lemma A.1 (Monotonicity under parametric maximization). *Let $\mathcal{X}$ be a nonempty set and let $f : \mathcal{X} \times \mathbb{R}_+ \rightarrow \mathbb{R}$. Define the value function*

$$V(\lambda) \equiv \sup_{x \in \mathcal{X}} f(x, \lambda).$$

If $f(x, \lambda)$ is weakly decreasing in λ for each fixed $x \in \mathcal{X}$, then $V(\lambda)$ is weakly decreasing in λ .

Proof. Let $0 \leq \lambda_1 < \lambda_2$. For any $x \in \mathcal{X}$, monotonicity of f in λ implies $f(x, \lambda_1) \geq f(x, \lambda_2)$. Taking the supremum over $x \in \mathcal{X}$ on both sides yields

$$\sup_{x \in \mathcal{X}} f(x, \lambda_1) \geq \sup_{x \in \mathcal{X}} f(x, \lambda_2),$$

i.e. $V(\lambda_1) \geq V(\lambda_2)$. Since $\lambda_1 < \lambda_2$ was arbitrary, V is weakly decreasing in λ . $\square$

Proof of Proposition 2.2. We conduct the proof in several steps:

Step 1: Formal definition of the market-level value function. For any market $s \in S$, blueprint quality $z \geq 0$, and organizational complexity level $C \geq 0$, define the market-level inner problem:

$$\Pi_s(z; C) \equiv \sup_{\substack{x \geq 0, \ell \geq 0, h \geq 0 \\ x + \ell \leq X(s)}} \{R(z_s, x) - q(C) \cdot h\}, \quad (19)$$

subject to the deployment constraint

$$z_s = \theta(A(h, \ell), d(s)) \cdot z. \quad (20)$$

Here, $R : \mathbb{R}_+ \times \mathbb{R}_+ \rightarrow \mathbb{R}$ is the market payoff function, $q : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is the unit cost of headquarters attention, $\theta : \mathbb{R}_+ \times \mathbb{R}_+ \rightarrow [0, 1]$ is the portability function, and $A : \mathbb{R}_+ \times \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is the adaptation production function. The blueprint creation cost $q(C)\chi(z)$ is common across markets and is handled at the outer level.

Step 2: Parametric dependence on organizational complexity. For fixed (z, s) , the value function $\Pi_s(z; C)$ depends on the complexity parameter C only through the unit cost $q(C)$, which multiplies the headquarters input h in the objective. All other primitives and constraints of the inner

problem— $R, \theta, A, d(s)$, and $\bar{X}(s)$ —as well as the feasible set, are independent of C . More precisely, we can rewrite (19) as

$$\Pi_s(z; C) = \sup_{(x, \ell, h) \in \mathcal{F}_s} \{R(\theta(A(h, \ell), d(s)) \cdot z, x) - q(C) \cdot h\},$$

where $\mathcal{F}_s \equiv \{(x, \ell, h) \in \mathbb{R}_+^3 : x + \ell \leq \bar{X}(s)\}$.

Step 3: Monotonicity of organizational complexity. By Assumption 2.1, $d(s) > 0$ for μ -almost every $s \in S$. Since $\mu(\Delta) > 0$ and $\Delta \subseteq S \setminus M$ is measurable, we have

$$C(M \cup \Delta) = \int_{M \cup \Delta} d(s) \mu(ds) = \int_M d(s) \mu(ds) + \int_\Delta d(s) \mu(ds) = C(M) + \int_\Delta d(s) \mu(ds).$$

Moreover, $d(s) > 0$ for μ -a.e. $s \in \Delta$ and $\mu(\Delta) > 0$, hence

$$\int_\Delta d(s) \mu(ds) > 0.$$

Therefore,

$$C(M \cup \Delta) > C(M). \tag{21}$$

Step 4: Monotonicity of the unit cost function. By Assumption 1.6 we have that $q(\cdot)$ is strictly increasing. Combined with (21), we obtain

$$q(C(M \cup \Delta)) > q(C(M)). \tag{22}$$

Step 5: Application of monotonicity under parametric maximization. Fix an incumbent market $s \in M$ and blueprint quality $z \geq 0$. For any feasible allocation $(x, \ell, h) \in \mathcal{F}_s$, define

$$f_{s,z}((x, \ell, h), \lambda) \equiv R(\theta(A(h, \ell), d(s)) \cdot z, x) - \lambda h, \quad \lambda \geq 0.$$

Define the parametric value function

$$V_{s,z}(\lambda) \equiv \sup_{(x, \ell, h) \in \mathcal{F}_s} f_{s,z}((x, \ell, h), \lambda).$$

By construction, the market-level value function satisfies

$$\Pi_s(z; C) = V_{s,z}(q(C)).$$

For each fixed $(x, \ell, h) \in \mathcal{F}_s$, the mapping $\lambda \mapsto f_{s,z}((x, \ell, h), \lambda)$ is weakly decreasing (strictly decreasing if $h > 0$, and constant if $h = 0$). Therefore, by Lemma A.1, the function $\lambda \mapsto V_{s,z}(\lambda)$ is weakly decreasing, and hence $\Pi_s(z; C)$ is weakly decreasing in $q(C)$.

Step 6: Conclusion. By Assumption 1.6 and $C(M \cup \Delta) > C(M)$ by (21), we have $q(C(M \cup \Delta)) > q(C(M))$ by (22).

By Step 5, the map $\lambda \mapsto \Pi_s(z; \cdot)$ evaluated at C such that $q(C) = \lambda$ is weakly decreasing in λ . Therefore,

$$\Pi_s(z; C(M \cup \Delta)) \leq \Pi_s(z; C(M)),$$

as claimed.

Step 7: Strictness of the inequality. Let $\lambda_1 \equiv q(C(M))$ and $\lambda_2 \equiv q(C(M \cup \Delta))$. Since $q(\cdot)$ is strictly increasing and $C(M \cup \Delta) > C(M)$, we have $\lambda_2 > \lambda_1$. The inequality is strict if and only if every optimal solution to the inner problem at λ_1 uses strictly positive headquarters attention.

Sufficiency. Suppose every optimal solution at λ_1 uses strictly positive headquarters attention. Let (x^*, l^*, h^*) be such an optimal allocation at λ_1 . Its value at the new higher cost λ_2 is strictly lower:

$$f_{s,z}((x^*, l^*, h^*), \lambda_2) < f_{s,z}((x^*, l^*, h^*), \lambda_1) = \Pi_s(z; C(M)).$$

Now consider any candidate allocation $(x, l, h) \in \mathcal{F}_s$ for the problem at λ_2 .

1. If $h > 0$, then $f_{s,z}((x, l, h), \lambda_2) < f_{s,z}((x, l, h), \lambda_1) \leq \Pi_s(z; C(M))$.
2. If $h = 0$, then $f_{s,z}((x, l, h), \lambda_2) = f_{s,z}((x, l, h), \lambda_1)$. However, because we assumed every optimal solution at λ_1 uses $h > 0$, any allocation with $h = 0$ must be strictly suboptimal at λ_1 . Therefore, $f_{s,z}((x, l, 0), \lambda_1) < \Pi_s(z; C(M))$.

In both cases, the value is strictly less than the original maximum. Since this holds for all feasible allocations, the new supremum must be strictly lower:

$$\Pi_s(z; C(M \cup \Delta)) < \Pi_s(z; C(M)).$$

Necessity. Conversely, if there exists an optimal solution $(\hat{x}, \hat{l}, \hat{h})$ at λ_1 with $\hat{h} = 0$, then

$$f_{s,z}((\hat{x}, \hat{l}, 0), \lambda_2) = f_{s,z}((\hat{x}, \hat{l}, 0), \lambda_1) = \Pi_s(z; C(M)).$$

Therefore $\Pi_s(z; C(M \cup \Delta)) \geq \Pi_s(z; C(M))$. Since Step 7 established the weak inequality $\leq$, it must be an equality, and strictness fails. $\square$

Proof of Proposition 2.3. We provide a rigorous proof establishing that the marginal value of blueprint quality aggregates across markets and is monotone in scope.

Step 1: Formalization of the firm's value function. Fix organizational complexity $C \geq 0$ and a served set $M \subseteq S$. The firm's total value from operating markets in M , as a function of blueprint quality $z \geq 0$, is given by

$$V(z; M, C) \equiv \int_{s \in M} \Pi_s(z; C) \mu(ds), \quad (23)$$

where $\Pi_s(z; C)$ is the market-level value function defined in Definition 2.

Step 2: Differentiability and the envelope theorem. Under regularity conditions (continuity and differentiability of R, θ, A), the market-level value function $\Pi_s(z; C)$ is differentiable in z for μ -almost every $s \in M$.

By the envelope theorem for parametric optimization, the derivative of $\Pi_s(z; C)$ with respect to z is given by

$$\frac{\partial \Pi_s(z; C)}{\partial z} = \frac{\partial R(z_s^*, x^*)}{\partial z_s} \cdot \frac{\partial z_s^*}{\partial z}, \quad (24)$$

where $(x^*, \ell^*, h^*) \in \arg \max$ solves the inner problem at (z, C) , and $z_s^* = \theta(A(h^*, \ell^*), d(s)) \cdot z$.

The envelope theorem states that we can ignore the direct effects of z on the choice variables (x, ℓ, h) when computing $\frac{\partial \Pi_s}{\partial z}$, because these variables are chosen optimally and the first-order conditions hold.

Step 3: Computing the marginal benefit at the market level. From the deployment constraint $z_s = \theta(a(s), d(s)) \cdot z$, we have

$$\frac{\partial z_s^*}{\partial z} = \theta(A(h^*, \ell^*), d(s)) \equiv \theta^*(s). \quad (25)$$

By Assumption 2.4 (blueprint effectiveness), $R(z_s, x)$ is weakly increasing in z_s , so

$$\frac{\partial R(z_s^*, x^*)}{\partial z_s} = R_1(z_s^*, x^*) \geq 0. \quad (26)$$

Combining (24), (25), and (26), we obtain

$$\frac{\partial \Pi_s(z; C)}{\partial z} = R_1(z_s^*, x^*) \cdot \theta^*(s) \geq 0. \quad (27)$$

This is the market-level marginal benefit of increasing blueprint quality.

Step 4: Aggregation across markets. The total marginal value of blueprint quality under served set

M is obtained by differentiating (23):

$$\frac{\partial V(z; M, C)}{\partial z} = \frac{\partial}{\partial z} \int_{s \in M} \Pi_s(z; C) \mu(ds). \quad (28)$$

We now establish conditions under which differentiation can be passed through the integral.

Lemma A.2 (Dominated convergence for derivatives). *Suppose:*

- (i) For each $s \in M$, $\Pi_s(z; C)$ is differentiable in z .
- (ii) There exists an integrable function $g \in L^1(M, \mu)$ such that for all z in a neighborhood of the evaluation point,

$$\left| \frac{\partial \Pi_s(z; C)}{\partial z} \right| \leq g(s) \quad \text{for } \mu\text{-a.e. } s \in M.$$

Then

$$\frac{\partial}{\partial z} \int_{s \in M} \Pi_s(z; C) \mu(ds) = \int_{s \in M} \frac{\partial \Pi_s(z; C)}{\partial z} \mu(ds).$$

Proof of Lemma A.2. This is a standard application of Lebesgue's dominated convergence theorem for differentiation under the integral sign. For any $\Delta z \rightarrow 0$,

$$\frac{V(z + \Delta z; M, C) - V(z; M, C)}{\Delta z} = \int_{s \in M} \frac{\Pi_s(z + \Delta z; C) - \Pi_s(z; C)}{\Delta z} \mu(ds).$$

By the mean value theorem, for each s ,

$$\frac{\Pi_s(z + \Delta z; C) - \Pi_s(z; C)}{\Delta z} = \frac{\partial \Pi_s(\xi_s; C)}{\partial z}$$

for some ξ_s between z and $z + \Delta z$. By condition (ii), this is bounded by $g(s) \in L^1(M, \mu)$. By dominated convergence, taking $\Delta z \rightarrow 0$ yields

$$\lim_{\Delta z \rightarrow 0} \frac{V(z + \Delta z; M, C) - V(z; M, C)}{\Delta z} = \int_{s \in M} \lim_{\Delta z \rightarrow 0} \frac{\Pi_s(z + \Delta z; C) - \Pi_s(z; C)}{\Delta z} \mu(ds) = \int_{s \in M} \frac{\partial \Pi_s(z; C)}{\partial z} \mu(ds) \quad \square$$

Step 5: Verifying the domination condition. We now show that condition (ii) of Lemma A.2 holds under our assumptions. From (27), the marginal benefit at market s is

$$\frac{\partial \Pi_s(z; C)}{\partial z} = R_1(z_s^*, x^*) \cdot \theta(A(h^*, \ell^*), d(s)).$$

By assumption, $\theta(a, d) \in (0, 1]$ for all (a, d) , so $\theta(A(h^*, \ell^*), d(s)) \leq 1$.

We need to bound $R_1(z_s^*, x^*)$. Since $z_s^* = \theta^*(s) \cdot z \leq z$ and $x^* \leq \bar{X}(s)$, if R has bounded derivatives on compact sets (which holds under regularity), there exists a function $\bar{g}(s)$ depending only on $\bar{X}(s)$ and z such that

$$|R_1(z_s^*, x^*)| \leq \bar{g}(s).$$

If $\bar{g}$ is integrable (which follows from Assumption 1.7: the integrability envelope $\bar{R} \in L^1(S, \mu)$ implies regularity of R sufficient for this), then condition (ii) holds with $g(s) = \bar{g}(s)$.

Step 6: Establishing monotonicity in scope. By Lemma A.2 and (27), we have

$$\frac{\partial V(z; M, C)}{\partial z} = \int_{s \in M} \frac{\partial \Pi_s(z; C)}{\partial z} \mu(ds) = \int_{s \in M} R_1(z_s^*, x^*) \cdot \theta^*(s) \mu(ds). \quad (29)$$

Now consider two served sets $M_1 \subseteq M_2 \subseteq S$. We have

$$\begin{aligned} \frac{\partial V(z; M_2, C)}{\partial z} &= \int_{s \in M_2} R_1(z_s^*, x^*) \cdot \theta^*(s) \mu(ds) \\ &= \int_{s \in M_1} R_1(z_s^*, x^*) \cdot \theta^*(s) \mu(ds) + \int_{s \in M_2 \setminus M_1} R_1(z_s^*, x^*) \cdot \theta^*(s) \mu(ds) \end{aligned} \quad (30)$$

By Assumption 2.4 (blueprint effectiveness), $R_1(z_s^*, x^*) \geq 0$ for all s , and $\theta^*(s) \geq 0$ by definition. Therefore,

$$\int_{s \in M_2 \setminus M_1} R_1(z_s^*, x^*) \cdot \theta^*(s) \mu(ds) \geq 0.$$

From (30), this implies

$$\frac{\partial V(z; M_2, C)}{\partial z} \geq \frac{\partial V(z; M_1, C)}{\partial z}. \quad (31)$$

Step 7: Economic interpretation and final statement. Equation (31) establishes the main result: the marginal return to blueprint quality is weakly increasing in the scope of operations. Because the blueprint is non-rival, an increase in z raises productivity in every market where it is deployed. Serving a larger set of markets $M_2 \supseteq M_1$ increases the number of markets that benefit from the marginal improvement, and since each market's marginal benefit is nonnegative (by Assumption 2.4 (blueprint effectiveness)), the aggregate marginal return increases.

Formally, the marginal return to blueprint quality is proportional to

$$\frac{\partial V(z; M, C)}{\partial z} = \int_{s \in M} R_1(z_s^*(s), x^*(s)) \cdot \theta(A(h^*(s), \ell^*(s)), d(s)) \mu(ds),$$

which is a sum (integral) of market-level marginal benefits, each weighted by the portability factor $\theta^*(s) \in (0, 1]$. This completes the proof. $\square$

Remark A.2 (Integrability and regularity). *The proof requires technical conditions ensuring that differentiation can be passed through the integral. These conditions are satisfied under standard regularity assumptions:*

- $R(z_s, x)$ is continuously differentiable with $R_1(z_s, x) \geq 0$ (Assumption 2.4),
- The derivative R_1 is bounded on compact sets,
- The integrability envelope $\bar{R} \in L^1(S, \mu)$ (Assumption 1.7) ensures sufficient regularity.

These assumptions are economically mild and standard in models with continua of markets.

Remark A.3 (Strict inequality). *The inequality in (31) is strict if and only if $M_2 \setminus M_1$ has positive measure and*

$$R_1(z_s^*, x^*) \cdot \theta^*(s) > 0$$

for a positive-measure subset of $M_2 \setminus M_1$. This holds generically when the additional markets contribute positive value and the blueprint has positive marginal product.

Proof of Proposition 2.4. Fix blueprint quality $z \geq 0$. With z fixed, the firm's choice of footprint $M \subseteq S$ solves

$$W(M) \equiv \int_{s \in M} \Pi_s(z; C(M)) \mu(ds) - q(C(M)) \chi(z), \quad (32)$$

where $C(M) \equiv \int_{s \in M} d(s) \mu(ds)$. By Proposition 2.2 (Organizational congestion), for each market s the function $\Pi_s(z; C)$ is weakly decreasing in C , and by Assumption 2.6 it is weakly decreasing in $d(s)$ for any fixed C .

Step 1 (No-crossing). Fix any $t \geq 0$ and define

$$A_t \equiv \{s \in M^* : d(s) > t\}, \quad B_t \equiv \{s \notin M^* : d(s) < t\}.$$

We claim that it is impossible that both $\mu(A_t) > 0$ and $\mu(B_t) > 0$.

Suppose instead that $\mu(A_t) > 0$ and $\mu(B_t) > 0$. Let $\varepsilon \equiv \min\{\mu(A_t), \mu(B_t)\} > 0$. By Assumption 1.1, the measure space (S, Σ, μ) is non-atomic, so there exist measurable subsets $A \subseteq A_t$ and $B \subseteq B_t$ with $\mu(A) = \mu(B) = \varepsilon$. Define the swapped footprint

$$M' \equiv (M^* \setminus A) \cup B.$$

Let $R \equiv M^* \setminus A$ denote the set of retained markets.

Complexity comparison. Since $d(s) > t$ on A and $d(s) < t$ on B , we have

$$C(M') = C(M^*) - \int_A d(s) \mu(ds) + \int_B d(s) \mu(ds).$$

Because $d(s) > t$ for all $s \in A$ and $\mu(A) = \varepsilon > 0$,

$$\int_A d(s) \mu(ds) > t \mu(A) = t\varepsilon.$$

Similarly, $d(s) < t$ for all $s \in B$ and $\mu(B) = \varepsilon$ imply

$$\int_B d(s) \mu(ds) < t \mu(B) = t\varepsilon.$$

Therefore $C(M') < C(M^*)$, and since $q(\cdot)$ is strictly increasing (Assumption 1.6),

$$q(C(M')) < q(C(M^*)). \quad (33)$$

Value comparison. Recall the firm's objective from (9):

$$W(M) = \int_{s \in M} \Pi_s(z; C(M)) \mu(ds) - q(C(M))\chi(z).$$

We compare $W(M')$ and $W(M^*)$ by decomposing each into contributions from retained markets, swapped markets, and the blueprint cost.

Write

$$W(M') = \int_R \Pi_s(z; C(M')) \mu(ds) + \int_B \Pi_s(z; C(M')) \mu(ds) - q(C(M'))\chi(z), \quad (34)$$

$$W(M^*) = \int_R \Pi_s(z; C(M^*)) \mu(ds) + \int_A \Pi_s(z; C(M^*)) \mu(ds) - q(C(M^*))\chi(z). \quad (35)$$

(i) *Retained markets.* By Proposition 2.2, $\Pi_s(z; \cdot)$ is weakly decreasing in organizational complexity. Since $C(M') < C(M^*)$, for every $s \in R$,

$$\Pi_s(z; C(M')) \geq \Pi_s(z; C(M^*)),$$

and hence

$$\int_R \Pi_s(z; C(M')) \mu(ds) \geq \int_R \Pi_s(z; C(M^*)) \mu(ds). \quad (36)$$

(ii) *Swapped markets.* Define

$$\underline{\pi}_B \equiv \operatorname{ess\,inf}_{s \in B} \Pi_s(z; C(M^*)), \quad \bar{\pi}_A \equiv \operatorname{ess\,sup}_{s \in A} \Pi_s(z; C(M^*)).$$

By construction, $d(s) < t$ for all $s \in B$ and $d(s) > t$ for all $s \in A$. Assumption 2.6 (heterogeneity ordering) implies that $\Pi_s(z; C)$ is weakly decreasing in $d(s)$ for any fixed C . Therefore $\underline{\pi}_B \geq \bar{\pi}_A$, and

$$\int_B \Pi_s(z; C(M^*)) \mu(ds) \geq \varepsilon \underline{\pi}_B \geq \varepsilon \bar{\pi}_A \geq \int_A \Pi_s(z; C(M^*)) \mu(ds). \quad (37)$$

Since $C(M') < C(M^*)$, Proposition 2.2 gives $\Pi_s(z; C(M')) \geq \Pi_s(z; C(M^*))$ for all $s \in B$. Thus

$$\int_B \Pi_s(z; C(M')) \mu(ds) \geq \int_B \Pi_s(z; C(M^*)) \mu(ds). \quad (38)$$

(iii) *Blueprint cost.* From (33),

$$-q(C(M'))\chi(z) \geq -q(C(M^*))\chi(z), \quad (39)$$

with strict inequality whenever $\chi(z) > 0$.

Conclusion of Step 1. Subtracting (35) from (34) and applying (36), (37), (38), and (39):

$$\begin{aligned} W(M') - W(M^*) &= \left[\int_R \Pi_s(z; C(M')) \mu(ds) - \int_R \Pi_s(z; C(M^*)) \mu(ds) \right] \\ &\quad + \left[\int_B \Pi_s(z; C(M')) \mu(ds) - \int_A \Pi_s(z; C(M^*)) \mu(ds) \right] \\ &\quad + [q(C(M^*)) - q(C(M'))] \chi(z) \\ &\geq 0 + \left[\int_B \Pi_s(z; C(M^*)) \mu(ds) - \int_A \Pi_s(z; C(M^*)) \mu(ds) \right] + [q(C(M^*)) - q(C(M'))] \chi(z) \\ &\geq 0 + 0 + 0 = 0. \end{aligned}$$

Moreover, the final term is strictly positive whenever $\chi(z) > 0$ (i.e., whenever $z > 0$), contradicting the optimality of M^* . Therefore, it is impossible that both $\mu(A_t) > 0$ and $\mu(B_t) > 0$.

Step 2 (Cutoff characterization). Define

$$d^* \equiv \operatorname{ess\,sup}_{s \in M^*} d(s) \in [0, \infty].$$

We show that $M^* = \{s : d(s) \leq d^*\}$ up to μ -null sets.

If $d(s) < d^*$, choose t such that $d(s) < t < d^*$. By definition of essential supremum, $\mu(\{u \in M^* : d(u) > t\}) > 0$. Step 1 then implies $\mu(\{u \notin M^* : d(u) < t\}) = 0$, and since $d(s) < t$, it follows that $s \in M^*$ μ -almost everywhere.

If $d(s) > d^*$, choose t with $d^* < t < d(s)$. Then $\mu(\{u \in M^* : d(u) > t\}) = 0$ by definition of d^* , which implies $M^* \subseteq \{d \leq t\}$ μ -almost everywhere. Hence $s \notin M^*$ μ -almost everywhere.

Combining these two cases yields

$$s \in M^* \iff d(s) \leq d^* \quad \mu\text{-a.e.},$$

establishing the cutoff structure of the optimal footprint. $\square$

A.3 Proofs for Section 2.3

Proof of Proposition 2.5. Fix blueprint quality $z \geq 0$, organizational complexity C , and a served set M . For each $s \in M$, the market-level inner problem (Definition 2) is

$$\Pi_s(z; C) \equiv \sup_{x \geq 0, \ell \geq 0, h \geq 0} \left\{ R(z_s, x) - q(C) h \right\} \quad \text{s.t.} \quad x + \ell \leq \bar{X}(s), \quad z_s = \theta(A(h, \ell), d(s)) z. \quad (40)$$

Fix two markets $s_1, s_2 \in M$ and write $d_i \equiv d(s_i)$ with $d_1 \leq d_2$. Let $h^*(s)$ denote an optimal (possibly set-valued) choice of headquarters attention in market s .

Step 1 (Reduce to an outer problem in h). For each market s and each $h \geq 0$, define the conditional value (optimizing out (x, ℓ) given h):

$$V(h; s) \equiv \sup_{x \geq 0, \ell \geq 0} \left\{ R(\theta(A(h, \ell), d(s)) z, x) \right\} \quad \text{s.t.} \quad x + \ell \leq \bar{X}(s). \quad (41)$$

Then the inner problem (40) is equivalent to

$$\Pi_s(z; C) = \sup_{h \geq 0} \left\{ V(h; s) - q(C) h \right\}. \quad (42)$$

Define the outer objective

$$\Phi(h; s) \equiv V(h; s) - q(C)h.$$

An optimizer $h^*(s)$ is any element of $\arg \max_{h \geq 0} \Phi(h; s)$.

Step 2 (Decreasing differences in (h, d)). We show that $\Phi(h; s)$ has decreasing differences in $(h, d(s))$, equivalently increasing differences in $(h, -d(s))$.

By Assumption 2.7 (adaptation regularity), the functions R , θ , and A are continuously differentiable with $R_1 \geq 0$, $\partial_a \theta \geq 0$, and $\partial_h A > 0$. By the envelope theorem applied to (41), at any h where a conditional optimizer $(x^*(s; h), \ell^*(s; h))$ is well-defined we have

$$\frac{\partial}{\partial h} V(h; s) = R_1(z_s(h), x^*(s; h)) \partial_a \theta(A(h, \ell^*(s; h)), d(s)) \partial_h A(h, \ell^*(s; h)) z, \quad (43)$$

where $z_s(h) = \theta(A(h, \ell^*(s; h)), d(s)) z$.

By Assumption 2.7, $R_1 \geq 0$ and $\partial_h A > 0$. Assumption 2.8 (single-crossing in heterogeneity) states that for any fixed $a \geq 0$, $\partial_a \theta(a, d)$ is weakly decreasing in d . This is the key channel generating the comparative static: holding the adaptation level a fixed, the marginal effectiveness of adaptation is lower in more heterogeneous markets.

To complete the argument, we require that this direct effect through $\partial_a \theta$ is not reversed by indirect effects operating through the optimal allocations (x^*, ℓ^*) or through R_1 .¹

Under this condition, holding h fixed, the marginal benefit of h is weakly lower in more heterogeneous markets:

$$d_1 \leq d_2 \implies \frac{\partial}{\partial h} V(h; s_1) \geq \frac{\partial}{\partial h} V(h; s_2) \quad \text{for all } h \geq 0. \quad (44)$$

Since $\Phi(h; s) = V(h; s) - q(C)h$ and $q(C)h$ does not depend on $d(s)$, the same property holds for Φ :

$$d_1 \leq d_2 \implies \frac{\partial}{\partial h} \Phi(h; s_1) \geq \frac{\partial}{\partial h} \Phi(h; s_2). \quad (45)$$

Condition (45) is exactly decreasing differences in (h, d) .

Step 3 (Monotone comparative statics). By Step 2, $\Phi(h; s)$ has increasing differences in $(h, -d(s))$.

¹This condition holds automatically when local capacity $\bar{X}(s)$ is constant across markets, so that optimal allocations (x^*, ℓ^*) depend on $d(s)$ only through the portability function θ . It also holds when R is separable in (z_s, x) or when R_1 is constant. More generally, the condition requires that Assumption 2.8 represents the dominant channel through which heterogeneity affects the return to headquarters attention.

Under standard regularity conditions ensuring existence of a maximizer of (42) (e.g. compactness of the feasible set for h or coercivity of Φ), the monotone comparative statics theorem (Milgrom–Shannon or Topkis) implies that the set of maximizers $\arg \max_{h \geq 0} \Phi(h; s)$ is nonincreasing in $d(s)$ with respect to the strong set order.

In particular, for any selections $h^*(s_i) \in \arg \max_{h \geq 0} \Phi(h; s_i)$,

$$d_1 \leq d_2 \implies h^*(s_1) \geq h^*(s_2).$$

This establishes the core–periphery allocation: headquarters attention is weakly decreasing in heterogeneity. $\square$

A.4 Proofs for Section 3

Proof of Lemma 3.1. Fix $\bar{C} < \infty$ and write $\bar{q} \equiv q(\bar{C}) < \infty$. Let $\bar{x} > 0$ and $\bar{h} > 0$ be as in Assumption 3.4, and let $\bar{\ell} \equiv 0$ and $\bar{\sigma} \equiv 0$.

Step 1 (Feasible near-core recipe). By Assumption 3.3(v), $\bar{x} + \bar{\ell} = \bar{x} \leq \inf_{s \in S} \bar{X}(s)$, hence $(x, \ell) = (\bar{x}, 0)$ is feasible in every market. The choice $(h, \sigma) = (\bar{h}, 0)$ is feasible by nonnegativity. Moreover, the standardization cost is zero under this recipe since $\kappa(0, z) = \psi(z)c(0) = 0$ by Assumption 3.3(iv).

Step 2 (Uniform deployment on near-core markets). Fix $\eta \in (0, 1)$ (take $\eta = \frac{1}{2}$). By Assumption 3.3(iii), $a \equiv A(\bar{h}, 0) > 0$. By Assumption 3.3(i)–(ii) and $g(0) = 1$, there exists $\varepsilon > 0$ such that for all $d \leq \varepsilon$,

$$\theta(a, d) \geq 1 - \eta.$$

Hence for any market s with $d_0(s) \leq \varepsilon$,

$$z_s = \theta(A(\bar{h}, 0), d_0(s)g(0))z = \theta(a, d_0(s))z \geq (1 - \eta)z = \frac{1}{2}z.$$

Step 3 (Uniform surplus lower bound for bounded congestion). Fix any $C \in [0, \bar{C}]$. Since $\hat{\Pi}_s(z; C)$ is a supremum over feasible choices, evaluating it at the feasible recipe $(\bar{x}, 0, \bar{h}, 0)$ yields, for any s with $d_0(s) \leq \varepsilon$,

$$\hat{\Pi}_s(z; C) \geq R(z_s, \bar{x}) - q(C)\bar{h} - \kappa(0, z) \geq R(\frac{1}{2}z, \bar{x}) - \bar{q}\bar{h}.$$

Define

$$\pi(z) \equiv R(\tfrac{1}{2}z, \bar{x}) - \bar{q}\bar{h}.$$

Then for all $z \geq 0$, all $C \in [0, \bar{C}]$, and all s with $d_0(s) \leq \varepsilon$, $\hat{\Pi}_s(z; C) \geq \pi(z)$.

Step 4 (Asymptotics). By Assumption 3.4(i), $R(z, \bar{x}) \rightarrow \infty$, hence $R(\tfrac{1}{2}z, \bar{x}) \rightarrow \infty$ and therefore $\pi(z) \rightarrow \infty$. Moreover, by Assumption 3.4(ii) (in the strengthened form with $\alpha = \tfrac{1}{2}$),

$$\frac{\pi(z)}{\chi(z)} = \frac{R(\tfrac{1}{2}z, \bar{x})}{\chi(z)} - \frac{\bar{q}\bar{h}}{\chi(z)} \rightarrow \infty,$$

since $\chi(z) \rightarrow \infty$ (because χ is increasing with $\chi(0) = 0$ and $z \rightarrow \infty$). Hence $\pi(z)/\chi(z) \rightarrow \infty$.

Step 5 (Positivity for large z). Since $\pi(z) \rightarrow \infty$, there exists $z_1 < \infty$ such that $\pi(z) > 0$ for all $z \geq z_1$. This establishes both claims. $\square$

Proof of Proposition 3.1. Let $\underline{\psi} \equiv \liminf_{z \downarrow 0} \psi(z) > 0$. Choose $\underline{z} > 0$ such that $\psi(z) \geq \underline{\psi}/2$ for all $z \in (0, \underline{z}]$. Fix $z \in (0, \underline{z}]$ and $C \geq 0$.

By definition of the extended market-level value function,

$$\hat{\Pi}_s(z; C) = \sup_{\substack{x \geq 0, \ell \geq 0, h \geq 0, \sigma \geq 0 \\ x + \ell \leq \bar{X}(s)}} \left\{ R(z_s, x) - q(C)h - \psi(z)c(\sigma) \right\} \quad \text{s.t.} \quad z_s = \theta(A(h, \ell), d_0(s)g(\sigma))z.$$

Since $\sigma = 0$ is feasible and $c(0) = 0$, we obtain the lower bound

$$\hat{\Pi}_s(z; C) \geq \sup_{\substack{x \geq 0, \ell \geq 0, h \geq 0 \\ x + \ell \leq \bar{X}(s)}} \left\{ R(\theta(A(h, \ell), d_0(s))z, x) - q(C)h \right\} = \Pi_s(z; C),$$

where the equality is exactly Definition 2 (the baseline market-level value function) evaluated at heterogeneity $d_0(s)$ (since $g(0) = 1$).

Conversely, for any feasible (x, ℓ, h, σ) in the extended problem,

$$R(z_s, x) - q(C)h - \psi(z)c(\sigma) \leq R(z_s, x) - q(C)h,$$

with strict inequality whenever $\sigma > 0$ (because $\psi(z) \geq \underline{\psi}/2 > 0$ and $c(\sigma) > 0$ for $\sigma > 0$). Taking suprema over feasible choices implies

$$\hat{\Pi}_s(z; C) \leq \Pi_s(z; C),$$

and thus $\hat{\Pi}_s(z; C) = \Pi_s(z; C)$ for all s and C , and any optimizer of the inner problem satisfies $\sigma^*(s) = 0$.

Therefore, in the outer problem the extended environment coincides with the baseline environment of Section 3 for all $z \in (0, \underline{z}]$. Applying Proposition 2.1 to the induced baseline problem yields bounded optimal complexity:

$$C(M^*(z), \sigma^*) = \int_{M^*(z)} d_0(s) g(\sigma^*(s)) ds = \int_{M^*(z)} d_0(s) ds < \infty.$$

□

Proof of Proposition 3.2. Fix any target scale $K < \infty$. Let $\bar{C} > 0$ be arbitrary and set $\bar{q} \equiv q(\bar{C}) < \infty$.

Step 1 (Uniform near-core surplus). By Lemma 3.1, there exist $\varepsilon > 0$, $z_1 < \infty$, and a function $\underline{\pi} : [z_1, \infty) \rightarrow (0, \infty)$ such that for all $z \geq z_1$, all $C \in [0, \bar{C}]$, and all markets s with $d_0(s) \leq \varepsilon$,

$$\hat{\Pi}_s(z; C) \geq \underline{\pi}(z) > 0.$$

Moreover, $\underline{\pi}(z) \rightarrow \infty$ and $\underline{\pi}(z)/\chi(z) \rightarrow \infty$ as $z \rightarrow \infty$.

Step 2 (Selecting K near-core markets). By Assumption 3.2, the set $S_\varepsilon = \{s \in S : d_0(s) \leq \varepsilon\}$ has infinite measure. Hence there exists a measurable subset $M_K \subseteq S_\varepsilon$ such that $\mu(M_K) = K$.

Step 3 (Bounding complexity). Define the standardization policy $\sigma(s) \equiv 0$ on M_K . Then $g(0) = 1$ implies

$$C(M_K, \sigma) = \int_{M_K} d_0(s) g(0) ds = \int_{M_K} d_0(s) ds \leq \varepsilon \mu(M_K) = \varepsilon K.$$

Choose $\bar{C} \equiv \varepsilon K$. Then $C(M_K, \sigma) \leq \bar{C}$ holds by construction.

Step 4 (Total payoff under bounded congestion). Consider the feasible firm plan that chooses blueprint quality $z \geq z_1$, market set M_K , and in each market implements an inner allocation that attains $\hat{\Pi}_s(z; C)$ up to an arbitrarily small $\delta > 0$ (possible by definition of the supremum). Since $C(M_K, \sigma) \leq \bar{C}$, we have $q(C(M_K, \sigma)) \leq q(\bar{C})$, and by Step 1,

$$\int_{M_K} \hat{\Pi}_s(z; C(M_K, \sigma)) ds \geq \int_{M_K} \underline{\pi}(z) ds = K \underline{\pi}(z).$$

The firm's total payoff (net of blueprint creation cost) therefore satisfies

$$V(z; M_K) \geq K \underline{\pi}(z) - q(\bar{C}) \chi(z).$$

Step 5 (Choosing z large enough). By Lemma 3.1(ii), $\underline{\pi}(z)/\chi(z) \rightarrow \infty$, hence there exists $z_K < \infty$ such that for all $z \geq z_K$,

$$K \underline{\pi}(z) > q(\bar{C}) \chi(z),$$

which implies $V(z; M_K) > 0$. This establishes that for all $z \geq z_K$ there exists a feasible plan with $\mu(M_K) = K$, complexity $C(M_K, \sigma) \leq \bar{C} < \infty$, and strictly positive total payoff. $\square$

Proof of Proposition 3.3. Define the value of the firm at blueprint quality z by

$$V(z) \equiv \sup_{(M, \sigma, x, \ell, h)} \left\{ \int_M \hat{\Pi}_s(z; C(M, \sigma)) ds - q(C(M, \sigma)) \chi(z) \right\},$$

where $C(M, \sigma) = \int_M d_0(s) g(\sigma(s)) ds$.

Step 1 (Bounded regime at low z). By Proposition 3.1, there exists $\underline{z} > 0$ such that for all $z \in (0, \underline{z}]$ the optimal standardization choice satisfies $\sigma^*(s) = 0$ for μ -a.e. $s \in M^*(z)$, and the extended problem coincides with the baseline problem of Section 3. Hence, by Proposition 3.1, any optimal solution has bounded complexity. In particular, since $g(0) = 1$,

$$C(M^*(z), \sigma^*) = \int_{M^*(z)} d_0(s) ds < \infty \quad \text{for all } z \in (0, \underline{z}].$$

This establishes the existence of a non-platform region at low blueprint quality.

Step 2 (Feasibility at arbitrary scale for large z). By Proposition 3.2, for every $K < \infty$ there exist $z_K < \infty$ and $\bar{C}_K < \infty$ such that for all $z \geq z_K$ the firm can achieve a feasible plan with $\mu(M) \geq K$, $C(M, \sigma) \leq \bar{C}_K$, and strictly positive total payoff. In particular, letting $K_n \rightarrow \infty$ be any diverging sequence, choose $z_n \geq z_{K_n}$ and corresponding feasible plans with $\mu(M_n) \geq K_n$ and positive payoff. This implies that for sufficiently large z platform-type expansion is feasible.

Step 3 (Definition of the threshold). Let

$$z^* \equiv \inf \left\{ z > 0 : \text{there exists a feasible plan with arbitrarily large footprint and strictly positive value} \right\}.$$

By Step 1 the set in braces is nonempty and bounded away from zero, hence $z^* > 0$. By Step 2 it

is nonempty, hence $z^* < \infty$.

Step 4 (Regime characterization). For any $z < z^*$, by definition of z^* no plan with arbitrarily large footprint can deliver positive value, so every optimal solution must remain in the bounded regime. For any $z > z^*$, by definition there exists a feasible plan that achieves arbitrarily large footprint with positive value; since the firm optimizes over all feasible plans, an optimal solution can be chosen to exhibit platform behavior. $\square$

A.5 Proofs for Section

Proof of Proposition 4.1. The optimal cutoff $d^*(z)$ satisfies the first-order condition equating the marginal benefit of serving the boundary market to its marginal cost:

$$z^\eta \theta(d^*(z)) = q(C(d^*(z))) \bar{h} + q'(C(d^*(z))) \bar{h} \cdot d^*(z) \cdot C'(d^*(z)), \quad (46)$$

where the left side is the revenue from the marginal market and the right side includes both the direct cost and the indirect congestion effect on inframarginal markets.

Consider two firms with blueprint qualities $z_H > z_L$. By condition (i), $d_H^* \equiv d^*(z_H) > d^*(z_L) \equiv d_L^*$. The firm with higher z serves more distant markets.

Now consider the marginal market for each firm:

- Firm L 's marginal market has heterogeneity d_L^* , with portability $\theta(d_L^*)$.
- Firm H 's marginal market has heterogeneity $d_H^* > d_L^*$, with portability $\theta(d_H^*) < \theta(d_L^*)$.

The key insight is that an increase in z has a smaller proportional effect on the marginal market condition for firm H than for firm L , because:

1. The marginal market for firm H has lower portability $\theta(d_H^*) < \theta(d_L^*)$, so the revenue gain from higher z is smaller.
2. The marginal market for firm H contributes more to complexity ($d_H^* > d_L^*$), so the cost of further expansion is higher.

Formally, implicitly differentiating (46):

$$\frac{d d^*}{d z} = \frac{\eta z^{\eta-1} \theta(d^*)}{-z^\eta \theta'(d^*) + (\text{cost terms increasing in } d^*)}. \quad (47)$$

As z increases and d^* expands:

- The numerator's $\theta(d^*)$ term shrinks (portability decay).
- The denominator's cost terms grow (higher d^* means higher complexity contribution).

Both effects reduce $\frac{dd^*}{dz}$ relative to d^*/z , implying that $\varepsilon_\Phi(z) = \frac{z}{d^*} \cdot \frac{dd^*}{dz}$ is decreasing in z . $\square$

Proof of Proposition 4.2. For a monotone transformation $\Phi(z)$ of a Pareto random variable z , the tail behavior of $\Phi(z)$ depends on how Φ behaves as $z \rightarrow \infty$.

Part (i): If $\Phi(z) \sim z^\gamma$ for large z with $\gamma < 1$ constant, then standard results imply the tail index of $\Phi(z)$ is $\alpha_z/\gamma > \alpha_z$. By Proposition 4.1, $\varepsilon_\Phi(z) < 1$ for large z , which is weaker than constant $\gamma < 1$ but still implies tail compression.

More precisely, let $\bar{\varepsilon} = \lim_{z \rightarrow z^*} \varepsilon_\Phi(z)$. If $\bar{\varepsilon} < 1$, then for z near z^* :

$$\Phi(z) \approx \Phi(z_0) \left(\frac{z}{z_0} \right)^{\bar{\varepsilon}}, \quad (48)$$

and the scope distribution inherits tail index approximately $\alpha_z/\bar{\varepsilon} > \alpha_z$.

Part (ii): Since $\varepsilon_\Phi(z) \rightarrow \bar{\varepsilon} < 1$ as $z \rightarrow z^*$, the scope function grows slower than any positive power of z with exponent 1. This implies the scope distribution's tail decays faster than any Pareto with index $\alpha_z - \varepsilon$.

Part (iii): If $\bar{d} < \infty$, then $\Phi(z) \leq \bar{d}$ for all $z < z^*$. The scope distribution therefore has support contained in $[0, \bar{d}]$, regardless of the unbounded support of z . $\square$